Contents

Chapter – 4

Introduction to Web Design and HTML

4.1 Core Concepts in Web Design

Internet

Internet is the connection of millions of computers worldwide that allows them to communicate and share information.

World Wide Web

World Wide Web is a system of sharing information over the Internet by using hypertext documents, which are linked together and can be accessed through a web browser.

Note: The Internet is the hardware, wires and protocols: the global infrastructure. The World Wide Web is a service, like email or online gaming, that runs on top of that infrastructure.

4.1.1 Basic Components of the Web

The web works through the combined use of three main components:

1. **HTML**: A markup language used to write and structure webpages.
2. **HTTP**: The protocol used to transfer webpages between web server and browser.
3. **Web Browser**: The software that receives, interprets and displays webpage content for the user.

Hypertext document file

A hypertext document is a type of document that contains links to other documents, allowing users to navigate between them easily.

- Like images this is a file type with extension .html or .htm.
- It is made by using HTML (HyperText Markup Language).

Webpage

A webpage is a single hypertext document or page that is displayed in a web browser. It is also called a web document. It can contain text, audio, video, still images, animation, links of other webpages and other multimedia content.

Website

A website is a collection of related webpages stored under a common domain name and connected through hyperlinks.

Web design

Web design is the process of planning, creating, arranging and presenting information on the World Wide Web. It includes the design of webpages, navigation system, content layout, links, images, and user interaction.

4.1.2 Basic Relationship

- A webpage is one single page.
- A website contains multiple related webpages.
- The first page shown after entering a website is called the **homepage**.
- Webpages are connected using hyperlinks.
- A web browser is used to view webpages.

Core Idea

$$\text{HTML} \rightarrow \text{Hypertext document files} \rightarrow \text{Webpage} \xrightarrow{\text{Web Design}} \text{Website} \rightarrow \text{World Wide Web} \rightarrow \text{Internet}$$

4.1.3 Advantages of Website

- Information can be published and accessed easily.
- A website can be accessed from anywhere in the world.
- It is available twenty-four hours a day.
- It reduces the need for physical communication.
- It supports text, images, audio, video and links.
- Information can be updated more easily than printed documents.
- It is useful for education, business, news, communication and entertainment.

4.2 Website

4.2.1 Website Structure

Website Structure

Website structure refers to the way webpages are organized, designed and interconnected within a website.

4.2.1.1 Common Pages in a Website

- **Home Page**: The first page displayed after entering a website. It works as the entry page or index page of the website and gives the main idea, purpose and links to important sections.
- **Main Section Page**: A page that represents a major topic or section of the website. It works like a menu or site index and links users to related subsection pages.
- **Subsection Page**: A page placed under a main section to present specific information. It should contain relevant content and navigation links to move through the website.

4.2.1.2 Types of Website Structure

4.2.1.2.1 Hierarchical Structure

- Pages are arranged like a tree.
- Starts from the homepage and branches into main sections and subsections.
- It is easy to manage and most commonly used.
- Users can easily understand their current position in the website.

4.2.1.2.2 Linear Structure

- Pages are connected in a fixed sequence.
- Users move step-by-step using links such as Next, Previous or Last.
- Suitable for tutorials, forms, sign-up pages or small documents.

4.2.1.2.3 Network Structure

- Each page is connected with many other pages.
- Users can move from one page to another without a fixed path.
- It gives flexibility, but may confuse users if navigation is not clear.

4.2.1.2.4 Hybrid Structure

- Combination of multiple structures.
- Usually combines hierarchical and linear or network structure.
- Most modern websites follow this model.
- Navigation tools such as menu, button and links should be clear and consistent.

4.2.2 Web Portal

Web Portal

A web portal is a website that collects important links, content and services from different sources in one place.

- It helps users access different information and services easily.
- Example: a government portal, education portal or service portal.

4.2.3 Types of Website

There are two main types of websites based on how their content works:

- Static Website
- Dynamic Website

4.2.3.1 Static Website

Static Website

A static website displays fixed content that does not change unless the source files are manually updated.

- Built mainly using HTML and CSS.

- Content remains fixed for all users.
- Users cannot directly input or update information.
- No database connection is normally required.
- Communication is mainly one-way: server to client.
- Loads quickly because pages are already prepared.

4.2.3.1.1 Advantages of Static Website

- Low development and hosting cost.
- Easy to develop, test and maintain.
- Loads quickly, even on slower connections.
- More secure because there is usually no database or server-side processing.
- Easier for search engines to index.

4.2.3.1.2 Disadvantages of Static Website

- Content cannot change automatically.
- User input or update is not supported.
- Updating content requires manual editing and re-uploading.
- Large static websites become difficult to control.
- Quality design can be time-consuming.

4.2.3.2 Dynamic Website

Dynamic Website

A dynamic website generates or changes content at runtime based on user input, request or database.

- Uses server-side scripting such as PHP, ASP.NET, JSP or similar technologies.
- Usually uses a database such as SQL or MySQL.
- Content can change for different users.
- Users can provide input and update information.
- Suitable for interactive services such as login, search, comment, online exam or e-commerce.

4.2.3.2.1 Advantages of Dynamic Website

- Content can be updated quickly.
- Can display specific pages or information for specific users.
- Handles large amounts of information using database.
- Supports user input and interactive features.
- Attractive and interactive layouts can be prepared.

4.2.3.2.2 Disadvantages of Dynamic Website

- Development and maintenance are more complex.
- Cost is higher than static websites.
- Loading may be slower because database and server processing are used.
- Security management is more important because input and database are involved.

4.2.3.3 Comparison between Static and Dynamic Website

Feature	Static Website	Dynamic Website
Content	Fixed content.	Content can change at runtime.
Loading Speed	Loads faster because pages are already prepared.	Loads slower because content may be generated dynamically.
Database	Does not usually use a database.	Usually uses a database to store data.
Security	More secure because there is less server-side processing.	Less secure than static if input and database are not properly protected.
User Input	User input or update is not supported.	User input and update are supported.
Technology	HTML and CSS are enough.	Requires HTML, CSS, server-side language and database.
Development	Easy to create and maintain.	Requires skilled developers and more control.
Update	Source files must be edited and uploaded again.	Content can be modified through server application or database.
Pages	Usually has a fixed number of pages.	Many pages can be generated dynamically.
Example Use	Simple information website.	E-commerce, online exam, login system or database website.

Table 4.1: Comparison between Static and Dynamic Website

4.2.4 Types of Webpage Based on Location

- **Local Webpage**: A webpage stored and opened from the user's own computer or local storage. It can be viewed without internet connection and without uploading it to a web server.
- **Remote Webpage**: A webpage stored on another computer or remote web server. It requires internet connection and a URL to access.

4.2.5 Website Classification by Purpose

- **Archive Site**: Stores old or important information for later use.
- **Business Site**: Presents information about a business, product or service.
- **E-commerce Site**: Supports buying and selling products or services online.
- **Community Site**: Connects people with common interests.
- **Database Site**: Provides searchable or organized data from a database.
- **Development Site**: Shares software, tools, documentation or project resources.
- **Directory Site**: Lists links, organizations, services or resources by category.
- **Download Site**: Provides files, software, documents or media for download.
- **Game Site**: Provides online games or gaming-related content.
- **Information Site**: Publishes subject-based information for readers.

- **News Site**: Publishes current news and updates.

4.2.6 Designing and Publishing a Website

4.2.6.1 Designing a Webpage

Web design is the creation of the external structure and front-end presentation of a website. A good website depends on layout, useful information, interface, linking, and up-to-date correct content.

4.2.6.2 Steps for Better Website Design

1. **Planning**: decide the objective of the website, target users, topics to publish and quality standard.
2. **Information architecture**: organize the website structure, page relationships and navigation.
3. **Layout design**: plan header, navigation, content, sidebar and footer.
4. **Content preparation**: prepare text, images, links, tables, forms and other materials.
5. **Implementation**: write HTML and CSS, then test the pages in a browser.
6. **Publishing**: upload the website files to hosting so users can access the website online.

4.2.6.3 Publishing a Website

Publishing means making the completed website available on the internet. To publish a website, the files are uploaded to a web hosting server and connected with a domain name.

- HTML, CSS, images and other files should be kept in organized folders.
- File names and folder paths should be correct because links and images depend on them.
- FTP or hosting control panels can be used to upload website files.
- After publishing, the website should be tested from a browser using its URL.

Summary

Design prepares the website plan and structure; publishing uploads the finished website so it can be accessed through the internet.

Summary

Static websites are simple and fast, whereas dynamic websites are flexible and interactive.

4.3 Web Browser and Search Engine

4.3.1 Web Browser

Web Browser

A web browser is application software used to access, retrieve, interpret and display webpages.

A browser works like an interpreter. It receives webpage code from the web server, converts it into visual output and shows the webpage on the screen.

4.3.1.1 Functions of Web Browser

- Opens websites and webpages using a URL.

- Interprets HTML and other webpage code.
- Displays text, images, links, forms and other webpage contents.
- Helps users move from one webpage to another through hyperlinks.
- Can download information or files from websites.
- Can upload information through forms or file upload systems.
- Can use search engines when the exact web address is not known.
- Example: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera.

4.3.2 Web Browsing

Web Browsing

Web browsing is the process of visiting and viewing webpages stored on different web servers through a web browser.

4.3.2.1 Methods of Web Browsing

- **Direct Browsing by URL**: The user types the known website address in the browser address bar and opens the webpage directly.
- **Browsing through Search Engine**: The user searches with keywords when the exact website address is not known.

4.3.3 Search Engine

Search Engine

A search engine is a tool that helps users find information on the internet by searching indexed webpages and showing relevant results.

A search engine collects and organizes webpage information. When a user searches with a word or phrase, it shows a list of related webpages.

4.3.3.1 Functions of Search Engine

- Helps users find information when the exact website address is unknown.
- Searches indexed webpages using keywords.
- Displays related webpage links as search results.
- Helps users reach websites quickly through the result list.
- Example: Google, Bing, Yahoo, DuckDuckGo, Ask, Yandex, Baidu, WolframAlpha (computational information-based search tool).

4.3.4 Difference Between Browser and Search Engine

Aspect	Web Browser	Search Engine
Meaning	Software used to open and display webpages.	Tool used to find information and webpages on the internet.

Aspect	Web Browser	Search Engine
Main Work	Reads webpage code and shows the webpage visually.	Searches indexed webpages and shows related results.
Use	Used when opening a website or viewing a webpage.	Used when the needed website or exact address is not known.
Input	URL or website address.	Keyword, phrase or question.
Examples	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera.	Google, Bing, Yahoo, DuckDuckGo, Ask.

Table 4.2: Difference between Web Browser and Search Engine

Summary

A web browser opens and displays webpages, while a search engine helps find webpages and information.

4.4 IP Address

IP Address

IP Address stands for Internet Protocol Address. It is a unique numerical address used to identify a device connected to a network on the internet.

Note: An IP address works like the digital address of a device. When sender sends data to a receiver, the receiver's IP address is used to ensure that the data reaches the correct destination.

4.4.1 Purpose of IP Address

- It identifies a device in a network.
- It helps send data to the correct destination.
- It allows communication between client and server.
- It is necessary for internet-based communication.

4.4.2 Types of IP Address

There are two main types of IP addresses:

- **IPv4**: 32-bit address, with four 8-bit octets, 1 octet = 8 bits, usually written in four decimal parts.
  Example: `11000000.10101000.00000001.00000001` $\xrightarrow{\text{in decimal}}$ `192.168.1.1`
- **IPv6**: 128-bit address, with eight 16-bit words, usually written in hexadecimal format.
  Example: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`

4.4.2.1 Network ID and Host ID

- The internet is a network of many networks.
- Each network has many hosts (computers, mobiles, servers etc.) in it.
- So each network and each device needs an unique identifier for communication.

- This is done by dividing the IP address into two parts: Network ID and Host ID.
  - → **Network ID**: identifies the network the device is connected to.
    Example: in `192.168.1.1`, the network ID is `192.168.1`.
  - → **Host ID**: identifies the specific device inside that network.
    Example: in `192.168.1.1`, the host ID is `1`.

Example 4.1

Q: Why is IP address required?

A: IP address is required because every device in a network must be uniquely identified. It helps data packets travel from the sender to the correct receiver.

Conclusion: Without IP addresses, devices on the internet could not communicate properly.

4.5 Domain Name and Domain Name System (DNS)

Domain Name

A **domain name** is a human-readable name used to identify a website on the internet.

Note: IP addresses are difficult for humans to remember. For this reason, domain names act as an alias for the numeric IP address.

DNS

DNS (Domain Name System) is the system that converts a readable domain name into its corresponding numeric IP address so the browser can load the resource.

The Bridge

- **IP Address** (192.168.1.1): used by computers in numeric form.
- **Domain Name** (*google.com*): used by humans in readable form.
- **DNS**: works like a phonebook that connects the readable name with the numeric address.

4.5.1 Some Technical Terms

- **FQDN** (Fully Qualified Domain Name): The complete domain name that identifies a specific host on the internet.
- **Subdomain**: An additional name placed before the main domain name, such as `www`.
- **Web Hosting**: The server space where website files are stored so users can access them through the internet.
- **Difference**: The *Domain* is your address; *Hosting* is the house where your data lives.

4.5.2 Domain Registration and ICANN

- **Domain Registration** is the process of reserving a domain name for use on the internet. It involves choosing a unique name and registering it with a domain registrar.

- Every registered domain name must be unique. If the same name could be registered by many people, users would not know which website to visit.
- For this reason, domain names are controlled through registration authorities.
- **ICANN** stands for "Internet Corporation for Assigned Names and Numbers".
- ICANN coordinates the global domain name and IP address system.
- Domain names are usually registered through domain registrars approved under this system.
- A registered domain gives a website a unique identity on the internet.

4.5.3 Types of Domain

- **Top Level Domain (TLD)** → The last part of a domain name. It shows the highest level/category of the domain.
  - → **gTLD (Generic TLD)** → General TLD, not tied to a country. Examples: `.com`, `.org`, `.net`
  - → **ccTLD (Country Code TLD)** → Country-based TLD. Examples: `.bd`, `.uk`, `.in`
- **Second Level Domain (SLD) / Main Domain Name** → The unique name chosen by the owner. It identifies the website or organization. Examples: `google`, `bbc`
- **Subdomain / Host Name** → A name placed before the main domain to identify a service or section. Examples: `www`, `mail`, `blog`

Example: Domain Structure

$$\underbrace{\texttt{mail}}_{\substack{\text{Subdomain}\\ \text{/ Host Name}}} \;.\; \underbrace{\texttt{google}}_{\substack{\text{Main Domain}\\ \text{/ SLD}}} \underbrace{\texttt{.com}}_{\substack{\text{TLD}\\ \text{gTLD}}}$$

$$\underbrace{\texttt{www}}_{\substack{\text{Subdomain}\\ \text{/ Host Name}}} \;.\; \underbrace{\texttt{bbc}}_{\substack{\text{Main Domain}\\ \text{/ SLD}}} \underbrace{\texttt{.co}}_{\substack{\text{Commercial}\\ \text{category}}} \underbrace{\texttt{.uk}}_{\substack{\text{TLD}\\ \text{ccTLD}}}$$

4.5.4 Common Top Level Domains / Generic TLDs

Domain	Used For
.com	Commercial organizations
.net	Network-related services
.edu	Educational institutions
.gov	Government organizations
.org	Non-profit organizations
.mil	Military organizations
.int	International organizations
.biz	Business organizations
.info	Information-based websites

Table 4.3: Common Top Level Domains

4.5.5 Country Code Top Level Domains

Code	Country	Code	Country
.bd	Bangladesh	.cn	China
.au	Australia	.fi	Finland
.in	India	.fr	France
.br	Brazil	.hk	Hong Kong
.uk	United Kingdom	.us	United States

Table 4.4: Common Country Code Top Level Domains

4.6 Web Hosting

Web Hosting

Web hosting means storing website files on an internet-connected server so that the website can be uploaded and accessed online.

- Hosting space is usually provided by an ISP or hosting company.
- Hosting cost depends on storage space, bandwidth, server resources and support.
- Different hosting services can be obtained from hosting providers by paying rent or service charge.
- Without hosting, a website's files cannot be publicly accessed through the internet.

4.6.1 Types of Web Hosting

4.6.1.1 Based on Platform or Operating System

- **Windows Hosting**: Hosting on a Windows server. It is mainly used for websites made with ASP/ASP.NET and Microsoft SQL Server.
- **Linux Hosting**: Hosting on a Linux server. It is mainly used for websites made with PHP and MySQL.

4.6.1.2 Based on Facilities

- **Free Hosting**: Basic hosting for small personal websites. It costs little or nothing, but bandwidth, security and domain facilities are limited.
- **Shared Hosting**: Many websites share the same server resources. It is cheaper, but speed, security, email, database and bandwidth are limited.
- **Dedicated Hosting**: One website or user gets a full server. It gives better speed, security, bandwidth, database support and control, so it is suitable for large websites.
  - → **Managed Hosting**: The hosting company manages setup, software, security, maintenance and control panel support. It is easier but more costly.
  - → **Unmanaged Hosting**: The website owner manages setup, software, security and maintenance. It is cheaper but needs technical skill.

- **Collocated Hosting**: The owner keeps their own server in the hosting company's data center and uses the company's power, network and support.

4.6.1.3 Email Services with Hosting

Hosting providers may also provide domain-based email accounts, such as `info@example.com`.

- **POP** (Post Office Protocol): downloads email from the server to a client device.
- **IMAP** (Internet Message Access Protocol): keeps email on the server and synchronizes it across devices.
- **Web-based Email / Webmail**: allows users to read and send email through a web browser.

4.6.2 Difference between Domain Name and Web Hosting

Domain Name	Web Hosting
The unique readable name of a website.	The server space where website files are stored.
Must be registered with a domain registrar.	Must be rented or configured from a hosting provider.
One domain name identifies one main web address.	One hosting account may contain one or many websites, depending on the plan.
Users cannot easily reach a website without a domain name or URL.	A website cannot be publicly available without hosting.
Domain registration does not require server configuration.	Hosting requires server configuration, storage, bandwidth and control panel settings.
Bandwidth is not the main issue in domain registration.	Bandwidth is an important issue in hosting.
The quality of the domain registrar is less visible to normal visitors.	The quality of the hosting provider affects speed, uptime, security and support.

Table 4.5: Difference between Domain Name and Web Hosting

4.7 URL (Uniform Resource Locator)

URL

A URL (Uniform Resource Locator) is the complete web address used to locate a resource on the internet.

Note: A URL identifies where a webpage or file is located and also tells the browser which protocol should be used to access it.

4.7.1 Components of URL

- **Protocol**: Defines the communication rule used to access the resource. Examples: `http://`, `https://`, `ftp://`
- **Domain Name**: Identifies the website or server. Example: `www.example.com`
- **Path**: Specifies the exact location of a file or page inside the server. Example: `/folder/page.html`

Structure of URL

$$\text{URL} = \text{Protocol} + \text{Domain Name} + \text{Path}$$

$$\text{URL} = \underbrace{\texttt{https://}}_{\text{Protocol}}\underbrace{\texttt{www.example.com}}_{\text{Domain Name}}\underbrace{\texttt{/folder/page.html}}_{\text{Path}}$$

Example: URL Structure

$$\underbrace{\texttt{https://}}_{\text{Protocol}}\ \underbrace{\texttt{www}}_{\substack{\text{Subdomain}\\ \text{/ Host Name}}}\ .\ \underbrace{\texttt{example}}_{\substack{\text{Main Domain}\\ \text{/ SLD}}}\ \underbrace{\texttt{.com}}_{\text{TLD}}\underbrace{\texttt{/folder/page.html}}_{\text{Path}}$$

$$\underbrace{\texttt{http://}}_{\text{Protocol}}\ \underbrace{\texttt{www}}_{\substack{\text{Subdomain}\\ \text{/ Host Name}}}\ .\underbrace{\texttt{corneliusenterprises}}_{\substack{\text{Main Domain}\\ \text{/ SLD}}}\ \underbrace{\texttt{.co}}_{\substack{\text{Commercial}\\ \text{category}}}\ \underbrace{\texttt{.uk}}_{\substack{\text{Country Code}\\ \text{TLD}}}$$

Summary

A URL gives both the access method and the location of a web resource.

4.8 Protocol

Protocol

A protocol is a set of rules that controls how data is transmitted between computers on the internet.

4.8.1 Common Internet Protocols

- **HTTP** (Hyper Text Transfer Protocol): Used to transfer webpages between browser and web server.
- **FTP** (File Transfer Protocol): Used to upload and download files through the internet.
- **IP** (Internet Protocol): Handles addressing and routing so data packets reach the correct destination.
- **TCP/IP** (Transmission Control Protocol / Internet Protocol): Main protocol system used for internet communication.
- **SMTP** (Simple Mail Transfer Protocol): Used for sending email.
- **POP** (Post Office Protocol): Used for receiving or downloading email from a mail server.
- **VoIP** (Voice over Internet Protocol): Used for voice communication through the internet.

4.8.2 HTTP

HTTP

HTTP stands for Hyper Text Transfer Protocol. It is the web protocol used to exchange data between a browser and a web server.

Note: HTTP is used when a user visits or browses a webpage. The browser sends a request to the web server, and the server sends the requested webpage or resource back to the browser.

4.8.2.1 Functions of HTTP

- Creates a connection between the browser and the server.
- Sends the browser's request to the server.
- Brings the requested webpage or resource from the server to the browser.
- Disconnects the browser and server after the request-response process.

4.8.3 FTP

FTP

FTP stands for File Transfer Protocol. It is a protocol used to exchange files between computers through the internet.

Note: FTP is mainly used to upload and download files between a local computer and a remote computer or server. It is commonly used to upload website files to a hosting server.

4.8.3.1 Functions of FTP

- Transfers website files, documents, books, images, software and other normal files.
- **Download**: Receives a file from a remote computer or server to the local computer.
- **Upload**: Sends a file from the local computer to a remote computer or server.
- FTP software such as **WS-FTP** and **Fetch** can be used for upload and download.

4.8.3.2 Types of FTP Access

- **Private Access**: Requires username and password to access server files.
- **Public Access**: Allows users to access selected files without private login.

HTTP vs FTP

HTTP is used to transfer hypertext/webpage files between browser and web server. FTP is used to transfer normal files such as documents, images, software and website files between computers.

4.9 Web Communication

Web Communication

Web communication is the process of exchanging data between a client browser and a web server through the internet.

Note: The World Wide Web works on top of the internet using a client-server system. The browser works as the client, and the web server stores and sends the requested webpage or resource.

4.9.1 Correct Overall Idea

- **Internet**: The global network infrastructure that connects computers and servers.
- **World Wide Web**: A service that runs on the internet and uses webpages, hyperlinks, browsers and web servers.
- **Website Developer**: Creates website files using HTML, CSS, images, scripts and other resources.

- **Web Hosting**: Stores the website files on an internet-connected server.
- **IP Address**: The numerical address of the hosting server.
- **Domain Name**: The human-readable name registered through a domain registrar.
- **DNS**: Connects the domain name with the hosting server's IP address.
- **URL**: The complete address used by the browser to request a specific webpage or file.
- **Protocol**: The rule used for communication, such as HTTP or HTTPS.

4.9.2 Client-Server System

- A website works using two main parts: **client** and **server**.
- **Client**: The user's computer or browser that requests a webpage.
- **Server**: The computer where webpages or website files are stored and protected.
- The client needs an internet connection and a web browser to request webpages.
- The web server uses HTTP/HTTPS to respond to the client's request and supply webpages.

4.9.3 Website Creation Side

1. The developer creates hypertext documents using **HTML**.
2. Other files such as CSS, images, audio, video and scripts may also be added.
3. These files are organized together as a website project or website folder.
4. The website folder is uploaded to a web hosting server.
5. The hosting server has an IP address.
6. A domain name is registered through a domain registrar.
7. DNS records connect the domain name with the hosting server's IP address.

4.9.4 User Access Side

1. The user enters a URL in the web browser.
2. The browser reads the protocol, domain name and path from the URL.
3. DNS converts the domain name into the server's IP address.
4. The browser connects to the correct web server using HTTP or HTTPS.
5. The browser sends an HTTP request for the specific webpage or file.
6. The server searches for the requested HTML document or resource.
7. The server sends the requested data back as an HTTP response.
8. The browser interprets the HTML and related files.
9. The webpage is displayed on the screen.

Web Working Flow

Developer creates website files → files are uploaded to hosting server → domain name is connected to server IP through DNS → user enters URL → browser requests the file → server responds → browser displays the webpage.

4.10 Difference between IP Address, Domain Name, DNS and URL

Aspect	IP Address	Domain Name	DNS	URL
Meaning	A unique numeric address used to identify a device or server on a network.	A human-readable name used to identify a website.	A system that converts a domain name into its corresponding IP address.	A complete web address used to locate a specific resource on the internet.
Main User	Computers and network devices.	Humans.	Browser and network system.	Browser and user.
Main Work	Finds the actual destination server or device.	Gives an easy readable name for a website.	Connects the domain name with the correct IP address.	Gives the access method and exact resource location.
Points To	A device or server.	A website name.	The IP address of a domain.	A webpage, file, image or resource.
Form	Numeric form.	Readable word form.	Background lookup system.	Full address with protocol, domain and path.
Example	`142.250.190.78`	`google.com`	`google.com` → `142.250.190.78`	`https://www.example.com/folder/page.html`

Table 4.6: Difference between IP Address, Domain Name, DNS and URL

4.11 HTML Fundamentals

HTML

HTML (HyperText Markup Language) is a markup language used to create webpages and web applications by describing how text, images, links, tables, forms and other contents should be organized and displayed in a browser.

HTML is not a programming language. It cannot process data, make decisions, or create a dynamic website by itself. It marks content with tags so that a browser can present that content as a webpage.

4.11.1 Concept of HTML

- HTML files are usually saved with the `.html` or `.htm` extension.
- A file created with HTML is generally known as a webpage or web document.
- HTML is platform independent; the same page can be opened in different operating systems and browsers.
- HTML can present text, audio, video, graphics, animation, hyperlinks, tables and attached contents.

- A conventional static webpage can be created with HTML, but dynamic or interactive behavior needs CSS, JavaScript, server-side scripts or databases.

4.11.2 Markup Language vs Programming Language

Markup Language	Programming Language
Uses tags to describe how content should be displayed.	Uses logic, variables, conditions and instructions to process data.
Cannot process data by itself.	Can process data and produce calculated results.
Cannot take and handle user input by itself.	Can take input, process it and produce output.
Example: HTML.	Example: C, JavaScript, Python, PHP.

Table 4.7: Difference between markup language and programming language

4.11.3 Short History of HTML

- Tim Berners-Lee invented HTML in 1990 while working at CERN in Geneva, Switzerland.
- HTML became widely known through the Mosaic browser developed by NCSA.
- HTML 2.0 was released in 1995.
- HTML 3.2 was released in January 1997.
- HTML 4.0 was released in December 1997.
- HTML5 is the modern major version; HTML 5.2 was released in December 2017.

4.11.4 Advantages of HTML

- It is a user-friendly open technology.
- It is used all over the world.
- Almost every web browser supports HTML documents.
- Its syntax is easy to learn and use.
- It can be written in any text editor.
- It is lightweight; normal HTML pages require less hosting space.
- It is not expensive and does not require a special license.

4.11.5 Limitations of HTML

- It can create static webpages but cannot create dynamic webpages by itself.
- It cannot process data or interact with a database.
- Large pages may require many lines of code.
- Security is limited because processing and validation are not handled by HTML alone.
- Complex websites require CSS, JavaScript and server-side technologies in addition to HTML.

Summary

HTML defines webpage structure. CSS controls presentation, JavaScript controls browser-side behavior, and server-side languages handle dynamic processing.

4.12 HTML Tags, Elements and Attributes

4.12.1 HTML Tag

HTML Tag

An HTML tag is a keyword enclosed in angle brackets that gives direction to the browser about how a content item should be structured or displayed.

Basic HTML Tag Syntax

```
<TagName> Content </TagName>
```

- The opening tag is also called the start tag. Example: `<p>`.
- The closing tag is also called the end tag. Example: `</p>`.
- The slash `/` is used in the closing tag.
- HTML tags are not case-sensitive, but lowercase tags are recommended by W3C.

4.12.2 Classification of HTML Tags

Tag Type	Meaning	Examples
Container tag	Has both opening and closing tags and contains content between them.	`<html>...</html>`, `<p>...</p>`
Empty tag	Has only an opening tag and no content between opening and closing tags.	` `, `<hr>`, `<img>`

Table 4.8: Container and empty HTML tags

In modern HTML examples, empty elements may be written with a closing slash, such as `<br/>` and `<hr/>`. Closing paired tags properly is important because missing end tags can produce unexpected output.

4.12.3 HTML Element

HTML Element

An HTML element is the complete unit formed by a start tag, element content and an end tag.

Element Structure

$$\text{Element} = \text{Start Tag} + \text{Content} + \text{End Tag}$$

For example, in `<p align="left">This is HTML class</p>`, the start tag is `<p align="left">`, the content is `This is HTML class`, and the end tag is `</p>`.

4.12.4 HTML Attribute

Attribute

An attribute is a special word written inside the start tag to give extra information or control the behavior, appearance or value of an HTML element.

Attribute Syntax

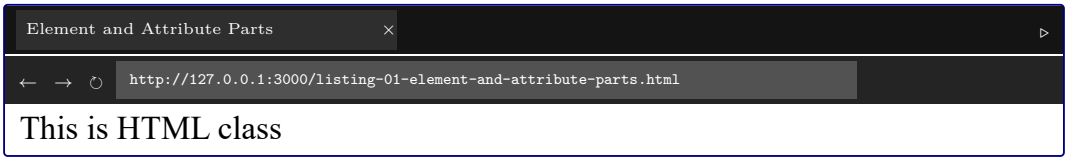
```
<TagName AttributeName="Attribute Value"> Content </TagName>
```

- An attribute has two parts: attribute name and attribute value.
- Attributes are written in the start tag, not in the end tag.
- Multiple attributes can be used in one tag.
- Lowercase attribute names are preferred.
- Attribute values are usually written inside double quotation marks.
- If an attribute value contains spaces, quotation marks are required.
- Single quotation marks can also be used when needed.

```html
<p align="left">This is HTML class</p>
```

Listing 4.1: Element and Attribute Parts

Browser Output



What to notice

- `align` is the attribute name.
- `left` is the attribute value.
- The attribute is written in the opening `p` tag.

4.12.5 HTML Comments

Comments are written for explanation inside source code. Browsers ignore comments, so they are not displayed on the webpage.

```html
<!-- This is a comment -->
```

Listing 4.2: HTML Comment

Browser Output

4.13 First HTML Document

HTML Document Structure

An HTML document is mainly formed with two sections inside the `html` element: the `head` section and the `body` section.

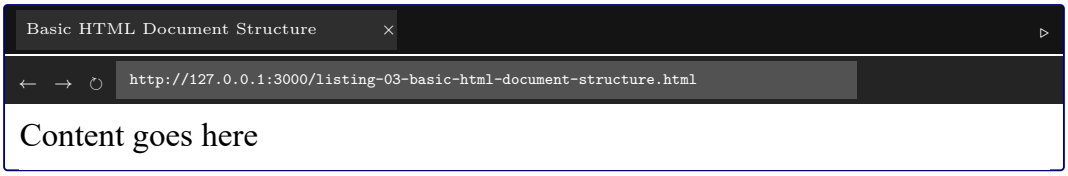
4.13.1 Basic Skeleton

This example shows the minimum structure of a valid HTML page. The `head` section stores information about the page, while the `body` section stores visible content.

```html
<!DOCTYPE html>
<html>
<head>
    <title>Page Title</title>
</head>
<body>
    Content goes here
</body>
</html>
```

Listing 4.3: Basic HTML Document Structure

Browser Output



Rendered output

Content goes here

4.13.2 Head Section

The head section contains document-related information such as document title, metadata, keywords, stylesheet links and scripts. This information is important for the browser and search engines, but most of it is not displayed as body content.

Element	Type	Use
`title`	Container	Sets the page title shown in the browser tab. A short useful title is preferred; no HTML tag should be written inside `title`.
`meta`	Empty	Stores document information such as charset, author, description or keywords.
`link`	Empty	Connects an external file such as a CSS stylesheet.
`base`	Empty	Sets a base URL for relative links in a document.
`style`	Container	Stores internal CSS rules.
`script`	Container	Stores or connects script commands.

Table 4.9: Common elements used in the HTML head section

4.13.3 Body Section

- The body section starts with `<body>` and ends with `</body>`.
- Text, headings, paragraphs, lists, tables, images, links and forms are placed inside `body`.
- Browsers display the visible content written inside the body section.

4.13.4 Block-Level and Inline Elements

Element Type	Behavior	Examples
Block-level element	Starts on a new line and usually takes the full available width.	`div`, `p`, `h1`, `ul`, `li`
Inline element	Does not start a new line; it stays inside the flow of text.	`a`, `b`, `span`, `strong`

Table 4.10: Block-level and inline HTML elements

4.13.5 Body Tag Attributes

The following old body attributes are important for exam/reference recognition. Modern webpages should use CSS instead.

Attribute	Use
`bgcolor`	Sets the page background color. Example: `<body bgcolor="green">`.
`background`	Sets an image as the page background. Example: `<body background="watermark.jpg">`.
`text`	Sets the color of normal page text.
`link`	Sets the color of unvisited hyperlinks.
`vlink`	Sets the color of visited hyperlinks.
`alink`	Sets the color of the active link while the mouse/cursor is on it.

Attribute	Use

Table 4.11: Old body attributes used in HTML

4.14 Creating and Running HTML Files

1. Open a simple text editor such as Notepad.
2. Write the HTML code.
3. Save the file using the `.html` or `.htm` extension.
4. Open the saved file in a web browser.
5. Edit the source file again when changes are needed, then refresh the browser.

Summary

HTML source code is written in a text editor, saved as an HTML file, and interpreted by a browser to display the webpage.

4.15 Headings, Paragraphs and Line Breaks

This lesson covers the basic tags used to organize text into headings, paragraphs, visible line breaks and simple separators. These tags are small, but they control how readable a webpage becomes.

In the table below, `-` means no special attribute or value is required for the basic use of that tag.

Tag / Entity	Attribute	Value	Meaning / Effect
`<h1>`–`<h6>`	-	-	Creates heading levels. `h1` is the largest/main heading and `h6` is the smallest heading.
`<p>`	-	-	Creates a paragraph block.
	`align`	`left`	Aligns paragraph text to the left.
		`center`	Aligns paragraph text to the center.
		`right`	Aligns paragraph text to the right.
		`justify`	Aligns paragraph text evenly on both left and right sides.
` `	-	-	Creates a manual line break inside text.
`<hr>`	-	-	Creates a horizontal line.
	`align`	`left`	Places the horizontal line on the left.
		`center`	Places the horizontal line in the center.
		`right`	Places the horizontal line on the right.
	`width`	pixel value	Sets a fixed line length, such as `200`.
		percentage value	Sets a relative line length, such as `60%`.
	`size`	pixel value	Sets the line thickness.

Tag / Entity	Attribute	Value	Meaning / Effect
	`color`	color name	Sets the line color using a named color, such as `blue`.
		color code	Sets the line color using a code, such as `#0000ff`.
` `	-	-	Creates one visible non-breaking space.

Table 4.12: Main tags, attributes and values for headings, paragraphs and line breaks

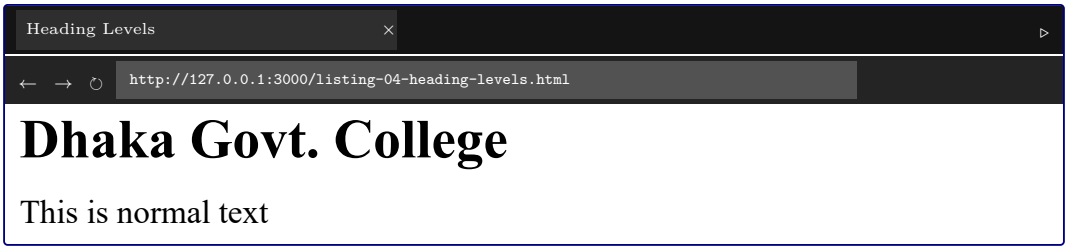
4.15.1 Heading Elements

HTML provides six heading levels from `h1` to `h6`. Headings are container elements and should be used to show document structure.

```
<h1>Dhaka Govt. College</h1>
<p>This is normal text</p>
<h6>Dhaka, Bangladesh</h6>
```

Listing 4.4: Heading Levels

Browser Output



4.15.2 Paragraph, Break, Horizontal Rule and Spacing

Normal paragraph text in HTML does not preserve every space, tab or Enter key from the source code. The browser reads the structure from tags, then displays the content according to HTML rules.

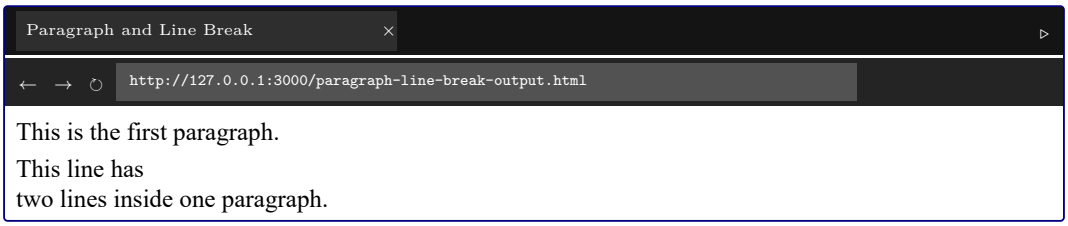
4.15.2.1 Basic Paragraph and Line Break

This example uses `p` for paragraph structure and `br` for manual line breaks. The paragraph tag creates a block of text, while `br` moves the next text to a new line without starting a new paragraph.

```
<p>This is the first paragraph.</p>
<p>This line has<br>two lines inside one paragraph.</p>
```

Listing 4.5: Paragraph and Line Break

Browser Output



What to notice

- A paragraph creates a separate block of text.
- `br` breaks the line at the exact position where it is written.
- Use `br` for real line breaks, not for full page layout spacing.

4.15.2.2 Horizontal Rule

The `hr` tag creates a horizontal line across the page. It is an empty tag, so it does not wrap any content.

```
<p>Section One</p>
<hr>
<p>Section Two</p>
```

Listing 4.6: Horizontal Rule

Browser Output

What to notice

- `hr` visually separates two parts of a page.
- It is not used for writing text.
- Modern webpages usually style horizontal lines with CSS.

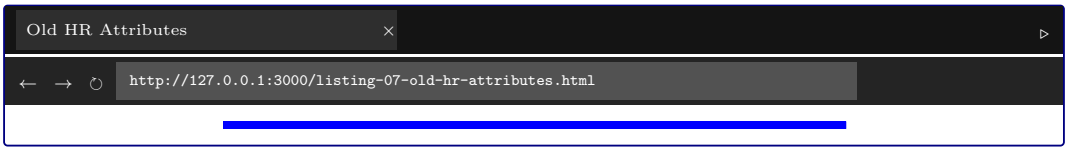
4.15.2.3 Old Horizontal Rule Attributes

Older HTML examples may use attributes inside `hr` to control alignment, width, thickness and color. These are exam/reference only, not modern practice.

```
<hr align="center" width="60%" size="4" color="blue">
```

Listing 4.7: Old HR Attributes

Browser Output



What to notice

- `align` controls the position of the line.
- `width` controls how long the line is.
- `size` controls line thickness.
- `color` controls line color.
- These old attributes may appear in exams, but CSS is preferred in modern HTML.

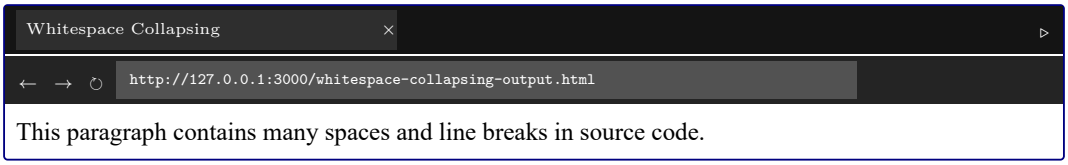
4.15.2.4 Whitespace Collapsing

Browsers collapse multiple normal spaces, tabs and source-code line breaks into a single space in normal paragraph text. Pressing Enter in the source file does not automatically create a visible new line in the browser.

```
<p>
This       paragraph
contains      many spaces
and line breaks in source code.
</p>
```

Listing 4.8: Whitespace Collapsing

Browser Output



What to notice

- Extra normal spaces usually display as one space.
- Source-code line breaks do not behave like `br`.
- Use proper tags, not random spacing, to control structure.

4.15.2.5 Non-breaking Space

The entity ` ` creates a visible space that the browser does not collapse. It is useful for small special spacing cases, but it should not be used as the main method for page layout.

```
<p>This   paragraph  has extra spaces.</p>
```

Listing 4.9: Using Non-breaking Space

Browser Output

What to notice

- ` ` is an HTML entity, not a tag.
- Each ` ` adds one visible non-breaking space.
- For layout spacing, CSS is the correct modern method.

4.16 Text Formatting in HTML

Formatting means giving proper shape and appearance to text on a webpage. Text formatting includes bold, italic, underline, big/small text, subscript, superscript, deleted text, inserted text, quotation and preformatted text.

Tag	Use	Example
`<b>`	Makes text bold.	`<b>Welcome</b>`
`<strong>`	Marks important text, usually shown bold.	`<strong>ICT</strong>`
`<i>`	Makes text italic.	`<i>Text</i>`

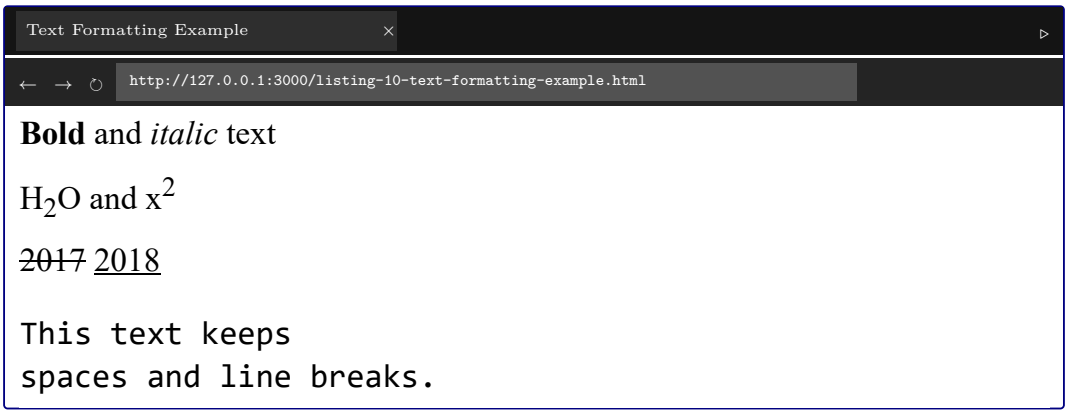
Tag	Use	Example
`<em>`	Emphasizes text, usually shown italic.	`<em>Text</em>`
`<u>`	Underlines text.	`<u>Text</u>`
`<ins>`	Shows inserted text, usually underlined.	`<ins>New</ins>`
`<big>`	Makes text bigger; exam/reference only.	`<big>Text</big>`
`<small>`	Makes text smaller.	`<small>Text</small>`
`<del>`	Shows deleted text.	`<del>Old</del>`
`<strike>`	Shows strikethrough text; exam/reference only.	`<strike>Old</strike>`
`<s>`	Shows text no longer correct; exam/reference recognition.	`<s>Old</s>`
`<sub>`	Shows subscript text.	`H<sub>2</sub>O`
`<sup>`	Shows superscript text.	`x<sup>2</sup>`
`<q>`	Shows inline quotation.	`<q>ICT</q>`
`<mark>`	Highlights/marks text.	`<mark>Important</mark>`
`<pre>`	Preserves spaces and line breaks.	`<pre>Text</pre>`

Table 4.13: Common text formatting tags

```html
<p><b>Bold</b> and <i>italic</i> text</p>
<p>H<sub>2</sub>O and x<sup>2</sup></p>
<p><del>2017</del> <ins>2018</ins></p>
<pre>
This text keeps
spaces and line breaks.
</pre>
```

Listing 4.10: Text Formatting Example

Browser Output



What to notice

- `b`, `i` and `u` mainly change visual appearance.
- `strong` and `em` also add meaning to the text.
- `sub` and `sup` are useful for formulas and powers.

- `pre` keeps source-code spacing and line breaks.

4.16.1 Visual vs Meaningful Formatting

Some tags only make text look different, while some tags also give meaning. For example, `b` and `strong` usually look similar, but `strong` means important text.

```
<p><b>Bold text</b></p>
<p><strong>Important text</strong></p>
<p><i>Italic text</i></p>
<p><em>Emphasized text</em></p>
```

Listing 4.11: Visual and Meaningful Formatting

Browser Output

What to notice

- `b` and `i` are visual formatting tags.
- `strong` and `em` are semantic formatting tags.
- In exam answers, mention both the output and the meaning where needed.

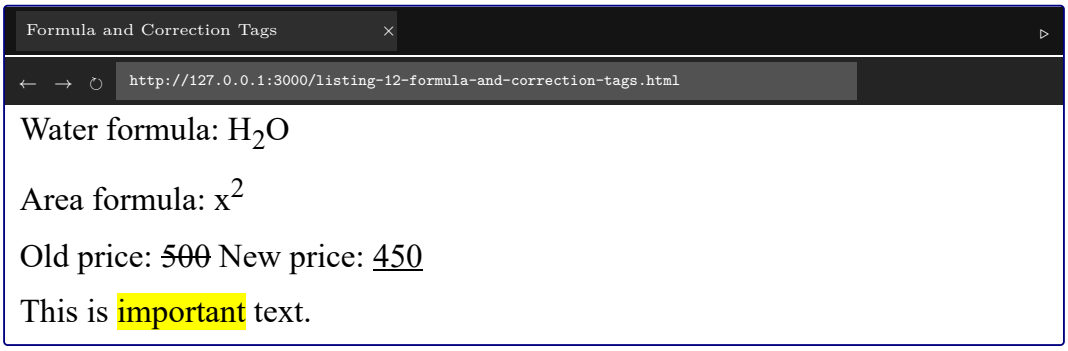
4.16.2 Formula and Correction Formatting

Subscript and superscript are important for science notation. Deleted and inserted text are useful when showing correction or update.

```
<p>Water formula: H<sub>2</sub>O</p>
<p>Area formula: x<sup>2</sup></p>
<p>Old price: <del>500</del> New price: <ins>450</ins></p>
<p>This is <mark>important</mark> text.</p>
```

Listing 4.12: Formula and Correction Tags

Browser Output



What to notice

- `sub` lowers text below the normal line.
- `sup` raises text above the normal line.
- `del` usually shows strikethrough text.
- `ins` usually shows underlined inserted text.

4.16.3 Old Text Tags

Some old tags may appear in textbooks or exam questions, but they are not modern practice. Modern HTML uses CSS for most visual text styling.

```html
<p><big>Big text</big></p>
<p><strike>Old strikethrough text</strike></p>
<p><s>Text no longer correct</s></p>
<center>This text is centered.</center>
```

Listing 4.13: Old Text Tags

Browser Output

What to notice

- `big`, `strike` and `center` are exam/reference only.
- Use CSS for modern font size, text alignment and decoration.

4.17 Font, Color, Style and Bangla Text

4.17.1 Font Tag

The old `font` tag was used to control font face, color and size. It is exam/reference only, not modern practice. Modern webpages should use CSS.

Old Font Tag Syntax

```html
<font face="fontname" color="fontcolor" size="number">Text</font>
```

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
	-	-	Displays the enclosed text using default browser style.
	`face`	font name	Sets the font family, such as `Arial`, `Verdana` or old Bangla font names.

`<font>`

Tag	Attribute	Value	Meaning / Effect
	`color`	color name	Sets text color using a named color, such as `red`.
		RGB value	Sets text color using RGB format, such as `rgb(255,0,0)`.
		hex code	Sets text color using hexadecimal format, such as `#FF0000`.
	`size`	`1`–`7`	Sets old HTML font size; `3` is the usual default.
		`+1`, `+2`	Makes text larger than the current/default size.
		`-1`, `-2`	Makes text smaller than the current/default size.

Table 4.14: Attributes of the old font tag

4.17.2 Color Values

- **Named colors**: black, gray, silver, white, yellow, lime, aqua, fuchsia, red, green, blue, purple, maroon, olive, navy, teal.
- **RGB colors**: Written as `rgb(red, green, blue)`, where each value is from 0 to 255. Example: `rgb(255,0,0)` means red.
- **Hexadecimal colors**: Written as `#RRGGBB`. Example: `#FF0000` means red and `#000000` means black.

Color	HEX	RGB
White	`#FFFFFF`	`rgb(255,255,255)`
Black	`#000000`	`rgb(0,0,0)`
Red	`#FF0000`	`rgb(255,0,0)`
Green	`#00FF00`	`rgb(0,255,0)`
Blue	`#0000FF`	`rgb(0,0,255)`
Yellow	`#FFFF00`	`rgb(255,255,0)`
Orange	`#FFA500`	`rgb(255,165,0)`
Violet	`#EE82EE`	`rgb(238,130,238)`

Table 4.15: Common HTML color values

4.17.3 Style Attribute

The `style` attribute is used for inline CSS. It can control exact font family, font size, font style and color.

```
<p style="font-family: Arial; font-size: 18px; color: red;">
    This is styled text.
</p>
```

Listing 4.14: Font Styling with CSS

Browser Output

4.17.4 Bangla Text in HTML

Bangla text may appear in older books using old font names such as `SutonnyMJ`. This is exam/reference only. The modern method is Unicode with UTF-8 encoding.

- Bangla can be written using old `font face` values, but this is not portable.
- Bangla can be shown using CSS `font-family`, but the font must exist on the user device.
- Unicode Bangla text with UTF-8 is the correct modern method.
- File names, image paths and link paths should be written in English characters for better compatibility.

```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Bangla Text</title>
</head>
<body>
    <p>Bangla Unicode text can be written here.</p>
</body>
</html>
```

Listing 4.15: UTF-8 and Unicode Bangla

Browser Output

4.18 Marquee

The `marquee` tag creates scrolling text in old HTML examples. It is exam/reference only, not modern practice.

Tag	Attribute	Value	Meaning / Effect
`<marquee>`	-	-	Creates scrolling text using default movement.
	`direction`	`left`	Moves text from right to left.
		`right`	Moves text from left to right.
		`up`	Moves text upward.

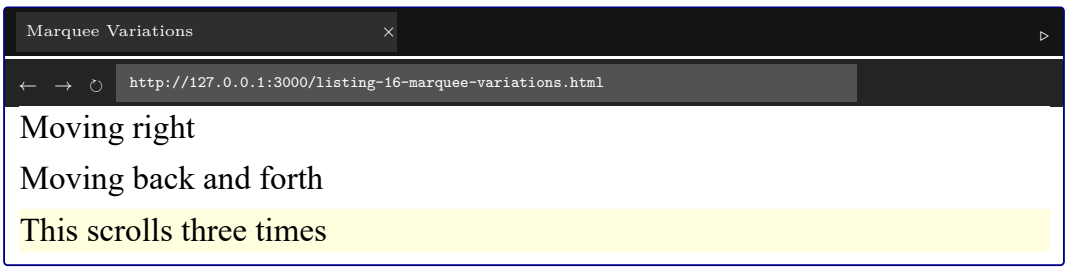
Tag	Attribute	Value	Meaning / Effect
		`down`	Moves text downward.
	`behavior`	`scroll`	Keeps scrolling continuously.
		`slide`	Moves once and then stops.
		`alternate`	Moves back and forth between sides.
	`scrollamount`	number	Controls movement speed; a larger number usually moves faster.
	`scrolldelay`	number	Controls delay between movements; a larger number usually moves slower.
	`loop`	number	Controls how many times the text scrolls.
	`bgcolor`	color value	Sets the marquee background color.

Table 4.16: Attributes of the old marquee tag

```html
<marquee direction="right">Moving right</marquee>
<marquee behavior="alternate" scrollamount="8">
    Moving back and forth
</marquee>
<marquee loop="3" scrolldelay="100" bgcolor="lightyellow">
    This scrolls three times
</marquee>
```

Listing 4.16: Marquee Variations

Browser Output



What to notice

- `direction` controls movement direction.
- `behavior="alternate"` makes the text bounce between sides.
- `loop`, `scrollamount` and `scrolldelay` control movement behavior.
- Modern pages should use CSS animation instead of `marquee`.

4.19 HTML Lists

HTML lists organize related items. The main list types are ordered list, unordered list and description list.

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<ol>`	-	-	Creates an ordered list with automatic numbering.
	`type`	`1`	Uses normal numbers: 1, 2, 3.
		`A`	Uses uppercase letters: A, B, C.
		`a`	Uses lowercase letters: a, b, c.
		`I`	Uses uppercase Roman numerals: I, II, III.
		`i`	Uses lowercase Roman numerals: i, ii, iii.
	`start`	number	Sets the starting number of the ordered list.
	`reversed`	`reversed`	Displays the ordered list in descending order.
`<ul>`	-	-	Creates an unordered list with bullet markers.
	`type`	`disc`	Shows filled circle bullets; old exam/reference style.
		`circle`	Shows hollow circle bullets; old exam/reference style.
		`square`	Shows square bullets; old exam/reference style.
`<li>`	-	-	Creates one item inside `ol`, `ul`, `dir` or old `menu`.
	`value`	number	Forces the number of a list item inside an ordered list.
`<dl>`	-	-	Creates a description list.
`<dt>`	-	-	Defines a term inside a description list.
`<dd>`	-	-	Defines the description of a term inside a description list.
`<dir>`	-	-	Old directory-style list tag; exam/reference only, not modern practice.
`<menu>`	-	-	Old menu-style list tag in textbook examples; exam/reference only, not modern practice.

Table 4.17: Main tags, attributes and values for HTML lists

4.19.1 Ordered List

An ordered list is used when the order of items matters. The browser automatically numbers the `li` elements.

```
<ol>
```

```html
    <li>Bangla</li>
    <li>English</li>
    <li>ICT</li>
</ol>
```

Listing 4.17: Ordered List

Browser Output

Ordered List

http://127.0.0.1:3000/listing-17-ordered-list.html

1. Bangla
2. English
3. ICT

4.19.1.1 Ordered List Variations

The `type` attribute changes the numbering style. The `start` attribute changes the starting number, and `value` can force the number of a specific list item.

```html
<ol type="A">
    <li>HTML</li>
    <li>CSS</li>
</ol>

<ol type="i" start="4">
    <li>Fourth item</li>
    <li>Fifth item</li>
</ol>

<ol>
    <li>Normal item</li>
    <li value="10">Forced number ten</li>
    <li>Continues as eleven</li>
</ol>
```

Listing 4.18: Ordered List Type and Start

Browser Output

Ordered List Type and Start

http://127.0.0.1:3000/listing-18-ordered-list-type-and-start.html

A. HTML
B. CSS

iv. Fourth item
v. Fifth item

1. Normal item
10. Forced number ten
11. Continues as eleven

What to notice

- Common `type` values are `1`, `A`, `a`, `I` and `i`.

- `start="4"` changes the first list number.
- `value="10"` changes only that item number and affects following items.

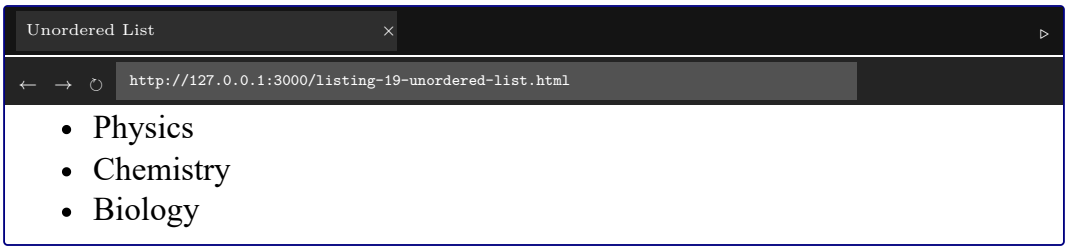
4.19.2 Unordered List

An unordered list is used when item order is not important. The browser normally displays bullet markers.

```html
<ul>
    <li>Physics</li>
    <li>Chemistry</li>
    <li>Biology</li>
</ul>
```

Listing 4.19: Unordered List

Browser Output



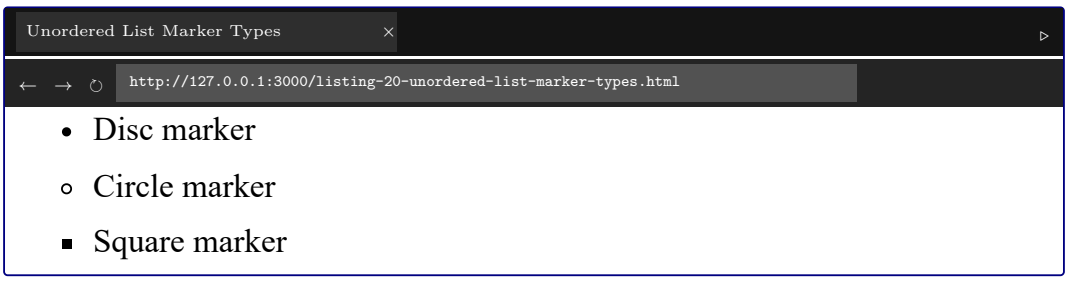
4.19.2.1 Unordered List Marker Variations

Older HTML examples may use `type` values such as `disc`, `circle` and `square`. Modern HTML should use CSS `list-style-type`.

```html
<ul type="disc">
    <li>Disc marker</li>
</ul>

<ul type="circle">
    <li>Circle marker</li>
</ul>

<ul type="square">
    <li>Square marker</li>
</ul>
```

Listing 4.20: Unordered List Marker Types

Browser Output



What to notice

- `type` on `ul` is old exam/reference style.
- CSS is the modern way to change list markers.

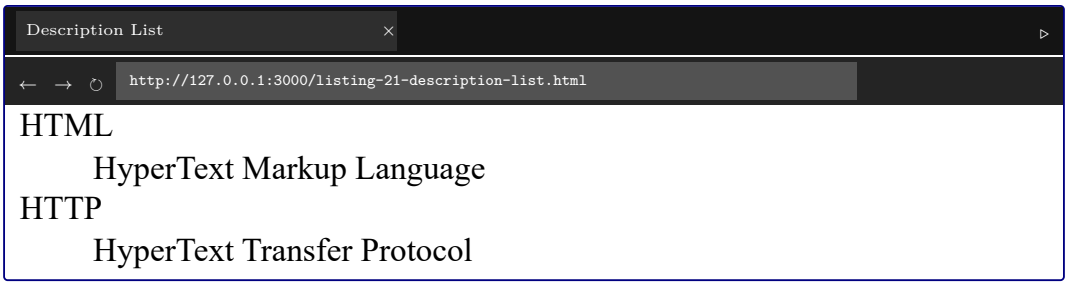
4.19.3 Description List

A description list is used for terms and their meanings. It uses `dl`, `dt` and `dd`.

```
<dl>
    <dt>HTML</dt>
    <dd>HyperText Markup Language</dd>
    <dt>HTTP</dt>
    <dd>HyperText Transfer Protocol</dd>
</dl>
```

Listing 4.21: Description List

Browser Output



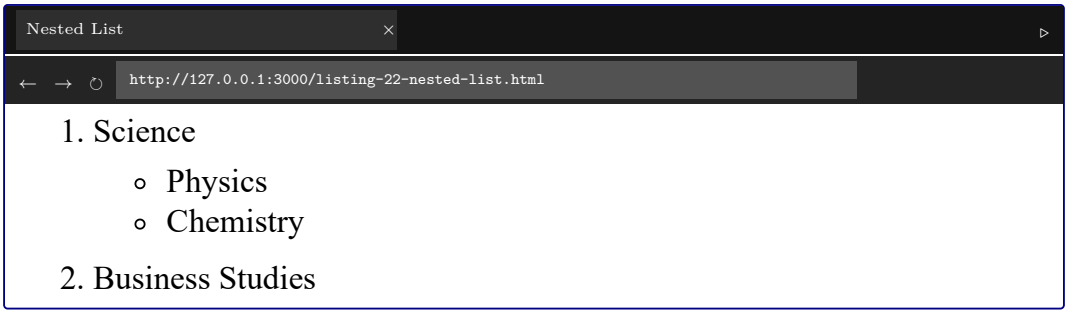
4.19.4 Nested List

A nested list means a list inside another list. It is useful for showing categories and subcategories.

```
<ol>
    <li>Science
        <ul>
            <li>Physics</li>
            <li>Chemistry</li>
        </ul>
    </li>
    <li>Business Studies</li>
</ol>
```

Listing 4.22: Nested List

Browser Output



4.19.5 Old Exam-Style List Tags

The old `dir` and `menu` tags may appear in old books. They are exam/reference only; use `ul`, `ol`, semantic navigation and CSS in modern pages.

```
<dir>
    <li>Old directory item</li>
```

```
</dir>

<menu>
    <li>Old menu item</li>
</menu>
```

Listing 4.23: Old List Tags

Browser Output

What to notice

- `dir` and old `menu` examples are not modern list practice.
- Use `ol` for sequence, `ul` for bullets and `dl` for term-description pairs.

Summary

Use `ol` for sequence, `ul` for bullets, and `dl` for term-description pairs.

4.20 Hyperlinks in HTML

Hyperlink

A hyperlink is a connection that allows a user to move from one webpage, file or resource to another.

Hyperlinks create bridges between pages. A website homepage can connect to other pages through hyperlinks. HTML pages, images, sound files, videos and documents can be referenced by hyperlinks.

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<a>`	-	-	Creates a hyperlink when used with a destination attribute.
	`href`	relative path	Links to a file using its location from the current page, such as `about.html`.
		absolute URL	Links to a complete web address, such as `https://www.example.com`.
		folder path	Links to a file inside another folder, such as `pages/about.html`.
		fragment ID	Links to a place inside a page, such as `#top`.
		`mailto:` address	Opens an email program when supported.

Tag	Attribute	Value	Meaning / Effect
		`tel:` number	Starts a phone call on supported devices.
	`target`	`_self`	Opens the link in the same tab or frame.
		`_blank`	Opens the link in a new tab or window.
		`_parent`	Opens the link in the parent frame.
		`_top`	Opens the link in the full browser window.
	`title`	text	Shows tooltip text when the mouse is placed over the link.
	`name`	anchor name	Old method for creating a bookmark point; modern pages usually use `id`.
	`download`	file name or empty	Tells the browser to download the linked file when supported.
`<link>`	`href`	file path or URL	Gives the location of the external resource.
	`rel`	`stylesheet`	Connects an external CSS file.
		`icon`	Connects a website icon.
	`type`	MIME type	States the resource type, such as `text/css`; often optional in HTML5.
`<base>`	`href`	base URL	Sets the base address for relative links in the page.
	`target`	`_self`	Opens relative links in the same tab by default.
		`_blank`	Opens relative links in a new tab/window by default.
		`_parent`	Opens relative links in the parent frame by default.
		`_top`	Opens relative links in the full browser window by default.
`<basefont>`	-	-	Old tag for setting default text style; exam/reference only, not modern practice.
	`face`	font name	Sets the default font face in old HTML examples.
	`size`	1–7	Sets the default font size in old HTML examples.
	`color`	color value	Sets the default text color in old HTML examples.

Table 4.18: Main tags, attributes and values for hyperlinks and head links

4.20.1 Anchor Tag

Hyperlink Syntax

```
<a href="url"> Link Text </a>
```

Two things are necessary to create a link: the destination address and the visible clickable text or object.

```
<a href="https://www.example.com">Visit Website</a>
```

Listing 4.24: Basic Anchor Link

Browser Output

What to notice

- `a` creates the hyperlink.
- `href` stores the destination address.
- The text between `<a>` and `</a>` becomes clickable.

4.20.2 Internal and External Links

- **Internal link**: Connects one page of a website with another page of the same website. Example: `studentinfo.html`.
- **External link**: Connects to a webpage or resource on another server or website. Example: `https://www.du.ac.bd/`.

4.20.3 Relative and Absolute Paths

Path Type	Meaning
Relative path	Gives a file location relative to the current HTML document. Example: `address.html` or `images/logo.jpg`.
Absolute path	Gives the complete address or fixed location. Example: `https://www.example.com/page.html`.

Table 4.19: Types of paths used in hyperlinks

4.20.4 Advantages of Hyperlink

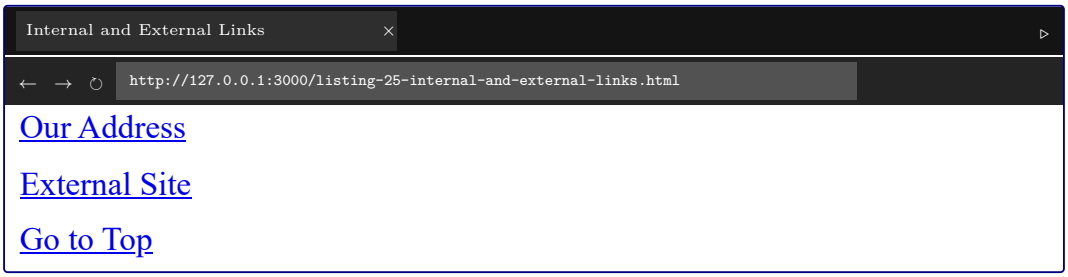
- Connects different pages of a website.
- Connects websites stored on different servers.

- Helps users find related information quickly.
- Saves time by allowing direct navigation.

```html
<p><a href="address.html">Our Address</a></p>
<p><a href="https://www.example.com" target="_blank">External Site</a></p>
<p><a href="#top">Go to Top</a></p>
```

Listing 4.25: Internal and External Links

Browser Output



4.20.5 Common Link Variations

Links can point to web addresses, local files, folders, email addresses, phone numbers or a specific place inside the same page.

```html
<a href="index.html">Same folder page</a>
<a href="pages/about.html">Page inside folder</a>
<a href="../home.html">Page in parent folder</a>
<a href="https://www.example.com">Absolute web link</a>
<a href="mailto:mail@example.com">Send email</a>
<a href="tel:+8801234567890">Call phone number</a>
```

Listing 4.26: Common Link Variations

Browser Output

What to notice

- Relative links depend on the current file location.
- Absolute links use the full address.
- `mailto:` opens an email program when supported.
- `tel:` is mainly useful on phones and supported devices.

4.20.6 Target and Download Attributes

The `target` attribute controls where a link opens. The `download` attribute tells the browser to download a file instead of opening it normally when supported.

```html
<a href="notice.html" target="_self">Open here</a>
<a href="https://www.example.com" target="_blank">
    Open in new tab
```

```
</a>
<a href="files/sample.pdf" download>Download PDF</a>
```

Listing 4.27: Target and Download Links

Browser Output

What to notice

- `_self` opens in the same tab.
- `_blank` opens in a new tab or window.
- `download` works best with local or same-site files.

4.20.7 Head Links: Link, Base and Basefont

The `link`, `base` and `basefont` tags are written in the `head` section. `basefont` is exam/reference only, not modern practice.

```
<head>
    <base href="https://www.example.com/">
    <link rel="stylesheet" href="style.css">
    <basefont face="Arial" size="4">
</head>
```

Listing 4.28: Head Link Tags

Browser Output

What to notice

- `link` connects external resources such as CSS files.
- `base` changes how relative links are resolved.
- `basefont` is old and should be replaced by CSS.

4.21 Images and Image Maps

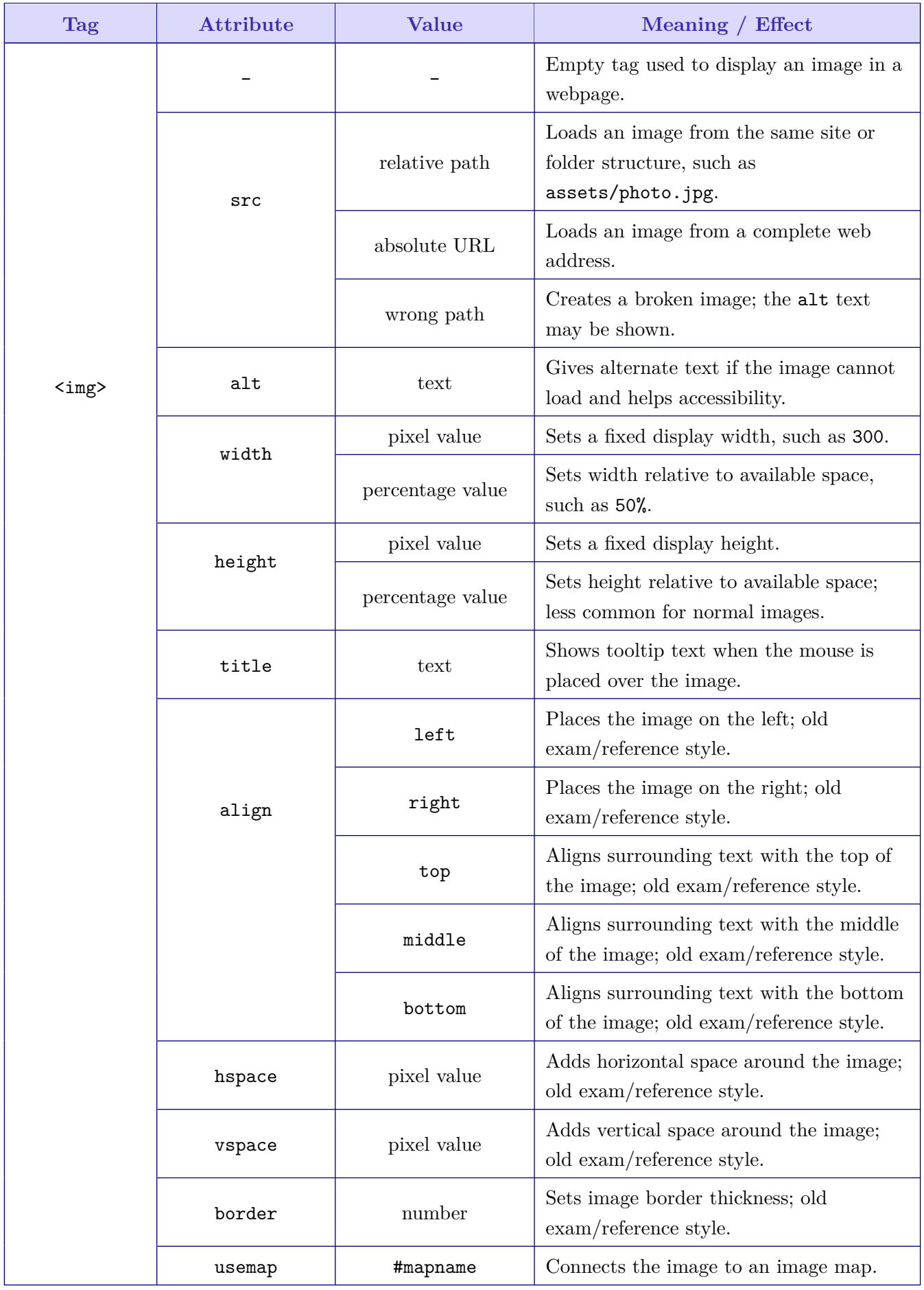
4.21.1 Adding Images

Image Tag

The `img` tag is an empty element used to add an image to a webpage.

Images, diagrams, charts, maps and graphics make webpages more attractive and can reduce the amount of text needed for explanation.

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
	-	-	Empty tag used to display an image in a webpage.
	src	relative path	Loads an image from the same site or folder structure, such as `assets/photo.jpg`.
		absolute URL	Loads an image from a complete web address.
		wrong path	Creates a broken image; the `alt` text may be shown.
	alt	text	Gives alternate text if the image cannot load and helps accessibility.
	width	pixel value	Sets a fixed display width, such as `300`.
		percentage value	Sets width relative to available space, such as `50%`.
	height	pixel value	Sets a fixed display height.
		percentage value	Sets height relative to available space; less common for normal images.
	title	text	Shows tooltip text when the mouse is placed over the image.
	align	left	Places the image on the left; old exam/reference style.
		right	Places the image on the right; old exam/reference style.
		top	Aligns surrounding text with the top of the image; old exam/reference style.
		middle	Aligns surrounding text with the middle of the image; old exam/reference style.
		bottom	Aligns surrounding text with the bottom of the image; old exam/reference style.
	hspace	pixel value	Adds horizontal space around the image; old exam/reference style.
	vspace	pixel value	Adds vertical space around the image; old exam/reference style.
	border	number	Sets image border thickness; old exam/reference style.
	usemap	#mapname	Connects the image to an image map.

Tag	Attribute	Value	Meaning / Effect
`<map>`	`name`	map name	Names the image map so an image can connect using `usemap`.
	`id`	map ID	Modern identifier for the map; often matches the `name`.
`<area>`	`-`	`-`	Empty tag used to define one clickable region inside an image map.
	`shape`	`rect`	Creates a rectangular clickable area.
		`circle`	Creates a circular clickable area.
		`poly`	Creates a polygon clickable area.
	`coords`	coordinate values	Gives the position and size of the clickable area.
	`href`	URL or path	Gives the destination for that clickable area.
	`alt`	text	Gives alternate text for the clickable area.
	`target`	target value	Controls where the linked area opens, such as `_blank`.

Table 4.20: Main tags, attributes and values for images and image maps

4.21.2 Advantages and Disadvantages of Web Graphics

Advantages	Disadvantages
Can reduce the amount of text.	Images take longer to download than text.
Attracts user attention.	Some browsers or devices may not support all graphics properly.
Can be used as graphical navigation links.	Too many images can make a page slow or heavy.

Table 4.21: Advantages and disadvantages of web graphics

4.21.3 Inline and External Images

- **Inline image**: Loads and displays directly with the webpage.
- **External image**: May not load immediately with the page and may require downloading or a separate source.

4.21.4 Image Syntax and Attributes

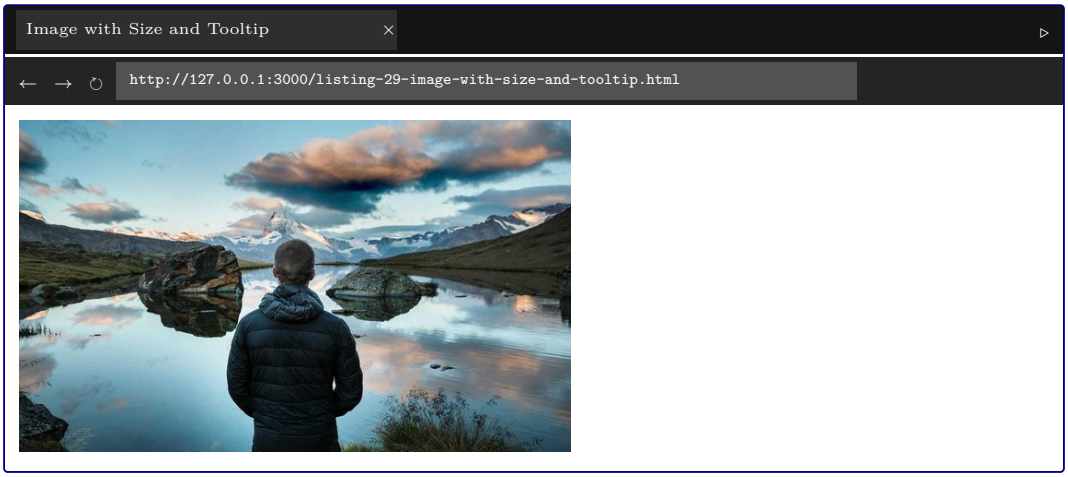
Image Syntax

```
<img src="url" alt="text">
```

```
<img src="assets/campus.jpg" alt="College campus"
     width="300" height="180" title="Campus Image">
```

Listing 4.29: Image with Size and Tooltip

Browser Output



What to notice

- `src` must point to the correct image path.
- `alt` appears when an image cannot load and helps accessibility.
- `width` and `height` control image display size.

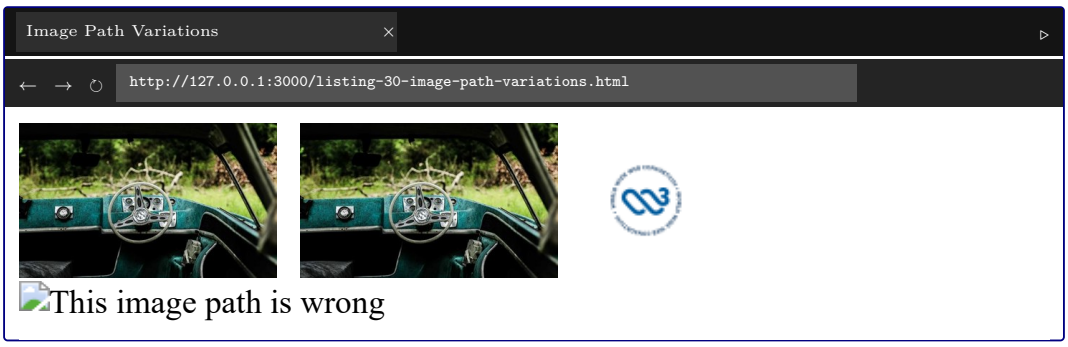
4.21.5 Image Paths and Broken Images

Image loading depends on the path written in `src`. If the path is wrong, the image will not load and the browser may show the alternate text.

```
<img src="assets/photo.jpg" alt="Local image" width="140">
<img src="assets/images/photo.jpg" alt="Folder image" width="140">
<img src="https://www.w3.org/Icons/w3c_home.png"
     alt="Online image" width="72">
<img src="wrong-name.jpg" alt="This image path is wrong" width="140">
```

Listing 4.30: Image Path Variations

Browser Output



What to notice

- File names and folder names must match exactly.
- A wrong `src` creates a broken image.
- The `alt` text is still meaningful when the image fails.

4.21.6 Old Image Spacing and Border Attributes

Older HTML examples may use image spacing and border attributes. These are exam/reference only; modern pages use CSS.

```html
<img src="assets/spacing-demo.png" alt="Spacing Demo"
     width="90" align="left" hspace="20" vspace="10" border="2">

<a href="home.html">
    <img src="assets/campus.jpg" alt="Home" width="120" border="0">
</a>
```

Listing 4.31: Old Image Attributes

Browser Output

What to notice

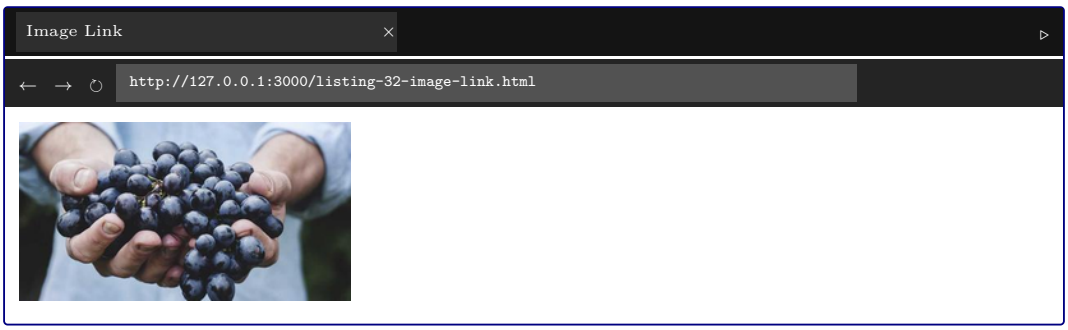
- `hspace` and `vspace` add space around the image.
- `border="0"` was often used to remove link image borders.
- CSS is preferred for alignment, spacing and borders.

4.21.7 Clickable Image

```html
<a href="https://www.educationboard.gov.bd">
    <img src="assets/link-image.png" alt="Education Board" width="180">
</a>
```

Listing 4.32: Image Link

Browser Output



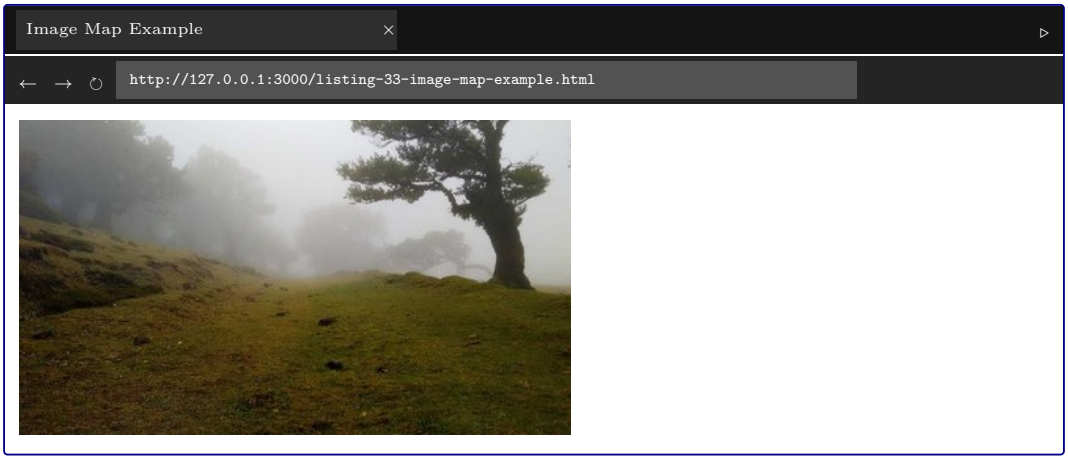
4.21.8 Image Map

An image map makes selected areas of one image clickable. It uses the `map` and `area` tags with coordinates.

```
<img src="assets/campus-map.jpg" alt="Campus Map"
     width="300" usemap="#campusmap">

<map name="campusmap">
    <area shape="rect" coords="10,20,120,90"
          href="library.html" alt="Library">
    <area shape="circle" coords="200,80,40"
          href="lab.html" alt="ICT Lab">
</map>
```

Listing 4.33: Image Map Example

Browser Output



Summary

Use `img` for normal images, image attributes for basic control, and `map`/`area` when selected parts of one image need separate links.

4.22 Media and Frames

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<audio>`	-	-	Adds audio content to a webpage.
	`controls`	-	Shows play, pause and volume controls.

Tag	Attribute	Value	Meaning / Effect
	autoplay	-	Starts playback automatically when allowed by the browser.
	loop	-	Repeats the audio after it finishes.
	muted	-	Starts the audio muted.
<video>	-	-	Adds video content to a webpage.
	controls	-	Shows video controls such as play, pause and volume.
	width	pixel value	Sets the display width of the video.
	height	pixel value	Sets the display height of the video.
	poster	image path	Shows an image before the video starts.
	autoplay	-	Starts playback automatically when allowed by the browser.
	loop	-	Repeats the video after it finishes.
	muted	-	Starts the video muted.
<source>	src	file path or URL	Gives the media file location.
	type	audio/wav	Identifies a WAV audio file.
		audio/mpeg	Identifies an MP3 audio file.
		video/mp4	Identifies an MP4 video file.
		video/webm	Identifies a WebM video file.
<iframe>	src	file path or URL	Gives the webpage or resource to display inside the frame.
	width	pixel or percentage value	Sets iframe width.
	height	pixel or percentage value	Sets iframe height.
	name	frame name	Names the iframe so links can target it.
	title	text	Describes the iframe for accessibility.
<frameset>	-	-	Old tag for dividing the browser window into frames; exam/reference only.
	rows	size list	Divides the page horizontally into row frames.
	cols	size list	Divides the page vertically into column frames.
<frame>	src	file path or URL	Gives the page loaded into that old frame.
	name	frame name	Names the old frame.

Tag	Attribute	Value	Meaning / Effect
	`scrolling`	`yes`, `no`, `auto`	Controls scrollbars in old frame examples.
	`noresize`	-	Prevents resizing of an old frame.
`<noframes>`	-	-	Shows fallback text when frames are not supported; exam/reference only.

Table 4.22: Main tags, attributes and values for media and frames

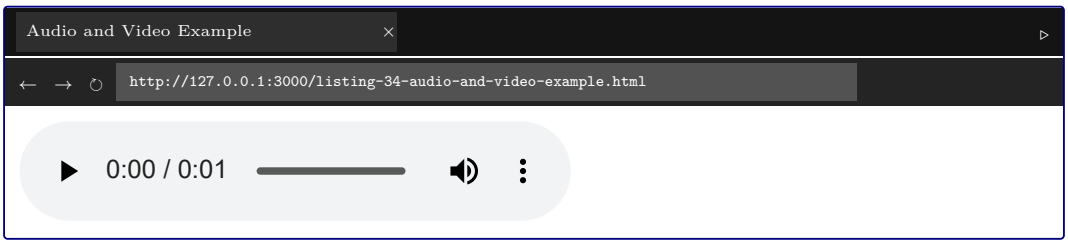
4.22.1 Audio and Video

Modern HTML uses `audio`, `video` and `source` tags to add media.

```html
<audio controls>
    <source src="assets/sample-audio.wav" type="audio/wav">
</audio>

<video controls width="320" height="180" poster="assets/video-poster.jpg">
    <source src="assets/sample-video.mp4" type="video/mp4">
</video>
```

Listing 4.34: Audio and Video Example

Browser Output



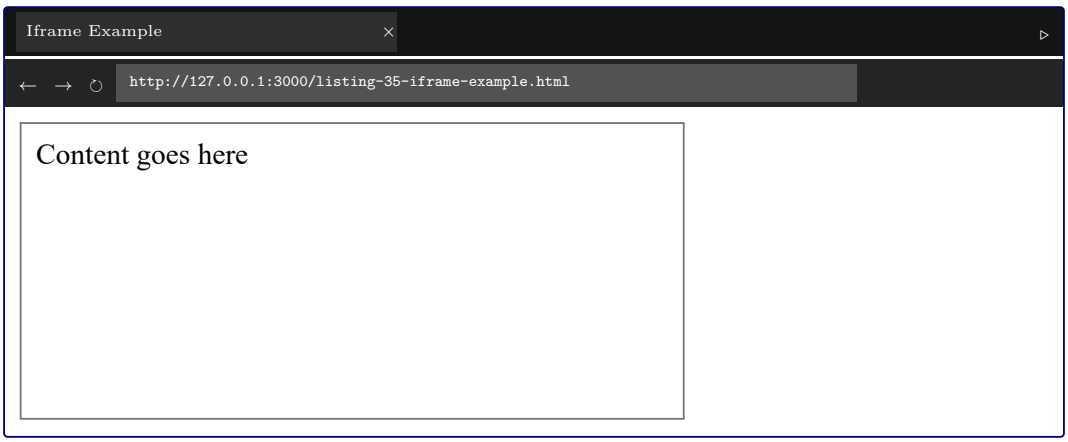
4.22.2 Iframe

The `iframe` tag displays another webpage or resource inside the current webpage. It is commonly used for embedded maps, videos or external pages.

```html
<iframe src="01-basic-skeleton.html" width="360" height="160">
</iframe>
```

Listing 4.35: Iframe Example

Browser Output



4.22.3 Old Frame Tags

Older HTML used `frameset`, `frame` and `noframes` to divide the browser window into separate frames. These are exam/reference only, not modern practice.

```html
<frameset cols="30%,70%">
    <frame src="menu.html">
    <frame src="content.html">
    <noframes>Your browser does not support frames.</noframes>
</frameset>
```

Listing 4.36: Old Frameset Example

Browser Output

What to notice

- `audio` and `video` need `controls` if users should play or pause media.
- `source` can provide the media file and type.
- `iframe` is still used, but old frame tags are only for exam/reference recognition.

4.23 HTML Tables

HTML Table

An HTML table presents information in rows and columns using the `table` element.

4.23.1 Parts of a Table

- **Caption**: title of the table.

- **Row**: horizontal line of cells.
- **Column**: vertical line of cells.
- **Cell**: box formed by the intersection of a row and a column.
- **Heading**: row or column title, usually shown in bold and centered.

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
<table>	-	-	Defines the whole table.
	border	number	Sets table border width; 0 means no visible border.
	width	pixel or percentage value	Sets table width.
	height	pixel or percentage value	Sets table height.
	align	left	Places the table on the left; old exam/reference style.
		center	Places the table in the center; old exam/reference style.
		right	Places the table on the right; old exam/reference style.
	cellspacing	pixel value	Sets space between table cells.
	cellpadding	pixel value	Sets space between cell border and cell content.
	bgcolor	color value	Sets table background color; old exam/reference style.
<caption>	-	-	Displays a title for the table.
<tr>	-	-	Creates one table row.
	align	left, center, right	Aligns row content horizontally; old exam/reference style.
		justify	Spreads row text evenly where supported; old exam/reference style.
		char	Aligns content around a character in old examples.
	bgcolor	color value	Sets background color for that row.
<td>	-	-	Creates a normal table data cell.
	colspan	number	Merges the cell across columns.
	rowspan	number	Merges the cell across rows.
	align	left	Aligns cell content to the left.
		center	Aligns cell content to the center.

Tag	Attribute	Value	Meaning / Effect
		right	Aligns cell content to the right.
	valign	top, middle, bottom	Aligns cell content vertically.
	bgcolor	color value	Sets background color for that cell.
<th>	-	-	Creates a heading cell; usually bold and centered.
	colspan	number	Merges the heading cell across columns.
	rowspan	number	Merges the heading cell across rows.
	align	left	Aligns heading content to the left.
		center	Aligns heading content to the center.
		right	Aligns heading content to the right.
	valign	top, middle, bottom	Aligns heading content vertically.
	bgcolor	color value	Sets background color for that heading cell.
<thead>	-	-	Groups the heading rows of a large table.
<tbody>	-	-	Groups the main data rows of a large table.
<tfoot>	-	-	Groups footer or summary rows of a large table.
<colgroup>	-	-	Groups columns so common column formatting can be applied.
<col>	span	number	Applies column settings to one or more columns.

Table 4.23: Main tags, attributes and values for HTML tables

4.23.2 Basic Table

```html
<table border="1">
    <caption>Student Info</caption>
    <tr>
        <th>Name</th>
        <th>Result</th>
    </tr>
    <tr>
        <td>Raza</td>
        <td>A-</td>
    </tr>
</table>
```

Listing 4.37: Basic HTML Table

Browser Output

What to notice

- `table` contains the whole table.
- `tr` creates one row.
- `th` creates heading cells.
- `td` creates normal data cells.

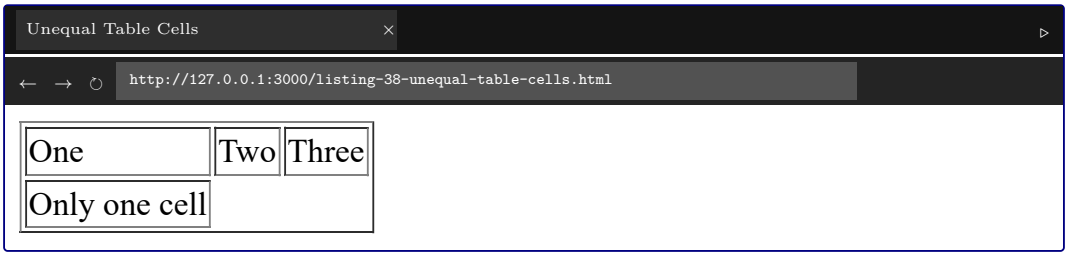
4.23.3 Unequal Number of Cells

If one row has more or fewer cells than another row, the browser still tries to draw the table. This can create an uneven table, so each row should usually follow the planned column count.

```
<table border="1">
    <tr>
        <td>One</td>
        <td>Two</td>
        <td>Three</td>
    </tr>
    <tr>
        <td>Only one cell</td>
    </tr>
</table>
```

Listing 4.38: Unequal Table Cells

Browser Output



What to notice

- The first row has three cells.
- The second row has only one cell.
- This is allowed by the browser, but it is usually bad table design unless merging is intended.

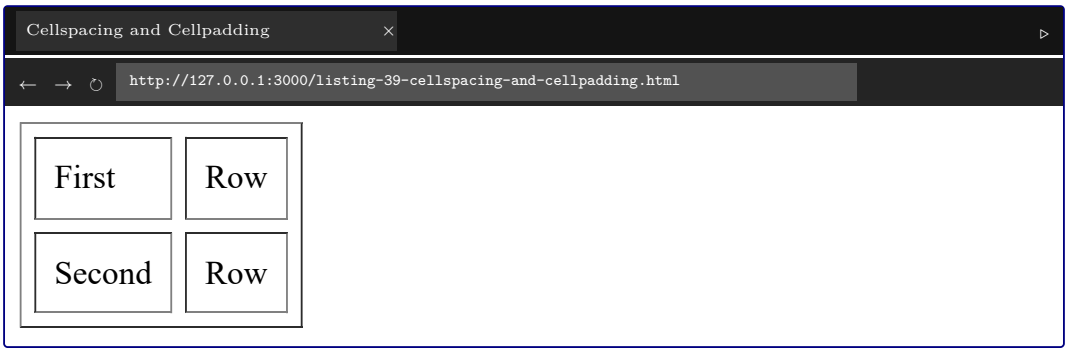
4.23.4 Cellspacing and Cellpadding

```
<table border="1" cellspacing="7" cellpadding="10">
    <tr>
```

```
        <td>First</td>
        <td>Row</td>
    </tr>
    <tr>
        <td>Second</td>
        <td>Row</td>
    </tr>
</table>
```

Listing 4.39: Cellspacing and Cellpadding

Browser Output



What to notice

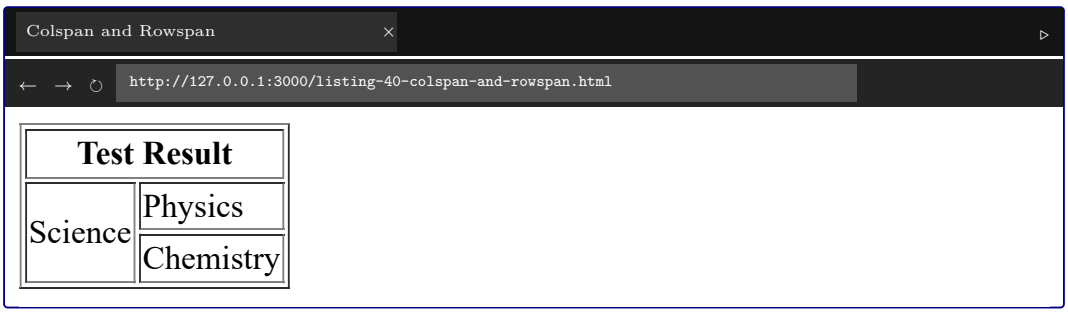
- `cellspacing` controls the gap between cells.
- `cellpadding` controls the gap inside each cell.
- Both are old HTML attributes; modern CSS uses `border-spacing` and `padding`.

4.23.5 Colspan and Rowspan

```
<table border="1">
    <tr>
        <th colspan="2">Test Result</th>
    </tr>
    <tr>
        <td rowspan="2">Science</td>
        <td>Physics</td>
    </tr>
    <tr>
        <td>Chemistry</td>
    </tr>
</table>
```

Listing 4.40: Colspan and Rowspan

Browser Output



What to notice

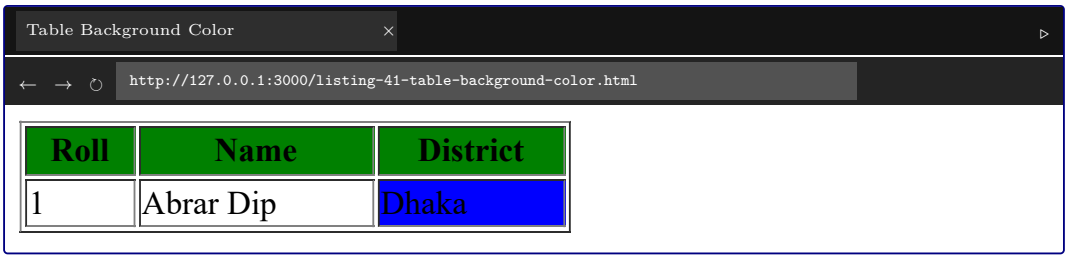
- `colspan="2"` makes one cell cover two columns.
- `rowspan="2"` makes one cell cover two rows.
- After merging, the number of remaining cells in that row or column must be adjusted.

4.23.6 Table Background Color

```html
<table border="1" width="300">
    <tr bgcolor="green">
        <th>Roll</th>
        <th>Name</th>
        <th>District</th>
    </tr>
    <tr>
        <td>1</td>
        <td>Abrar Dip</td>
        <td bgcolor="blue">Dhaka</td>
    </tr>
</table>
```

Listing 4.41: Table Background Color

Browser Output



4.23.7 Table Alignment and Empty Cells

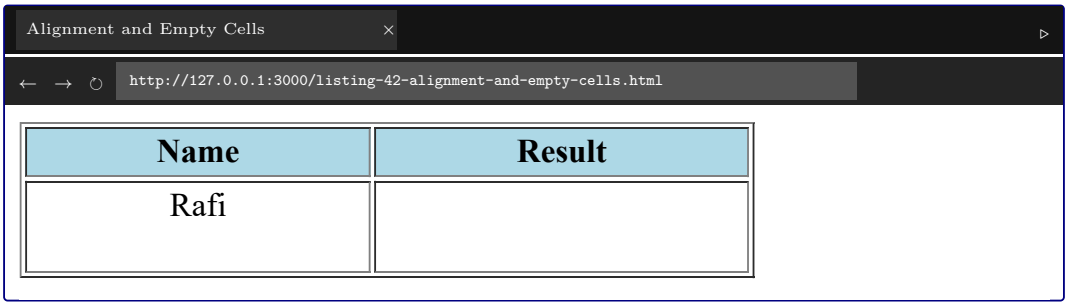
Old HTML examples may use `align`, `valign`, `bgcolor`, `width`, `height` and ` ` inside tables. These are important for exam recognition, but CSS is preferred in modern webpages.

```html
<table border="1" width="400" align="center">
    <tr bgcolor="lightblue">
        <th>Name</th>
        <th>Result</th>
    </tr>
    <tr height="50" align="center">
        <td valign="top">Rafi</td>
        <td> </td>
```

```
    </tr>
</table>
```

Listing 4.42: Alignment and Empty Cells

Browser Output



What to notice

- `align` controls horizontal alignment.
- `valign` controls vertical alignment inside a cell.
- ` ` keeps an otherwise empty cell visible in old examples.
- `bgcolor`, `width` and `height` are exam/reference style attributes.

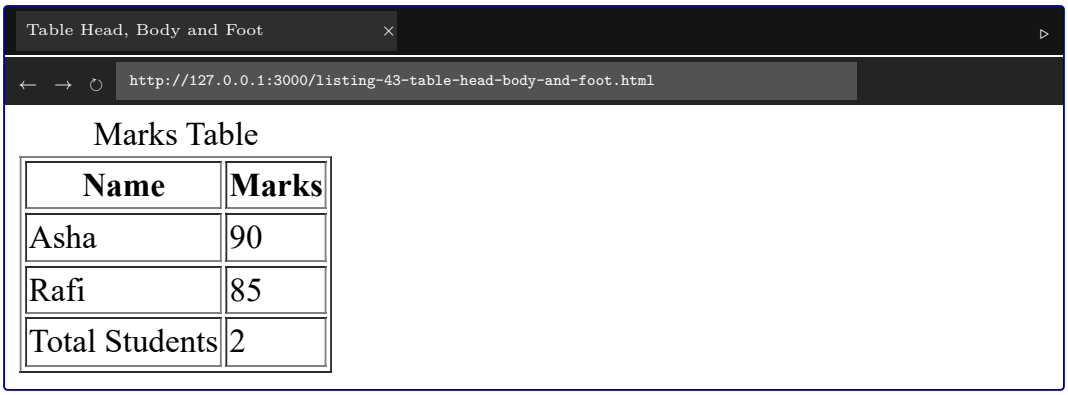
4.23.8 Table Sections

Large tables can be divided into `thead`, `tbody` and `tfoot`. This makes the structure clearer.

```
<table border="1">
    <caption>Marks Table</caption>
    <thead>
        <tr><th>Name</th><th>Marks</th></tr>
    </thead>
    <tbody>
        <tr><td>Asha</td><td>90</td></tr>
        <tr><td>Rafi</td><td>85</td></tr>
    </tbody>
    <tfoot>
        <tr><td>Total Students</td><td>2</td></tr>
    </tfoot>
</table>
```

Listing 4.43: Table Head, Body and Foot

Browser Output



What to notice

- `thead` contains heading rows.
- `tbody` contains the main table data.
- `tfoot` contains footer or summary rows.

Summary

For exams, know `table`, `tr`, `td`, `th`, `caption`, `border`, `cellspacing`, `cellpadding`, `colspan`, `rowspan`, `align`, `valign` and `bgcolor`.

4.24 Containers, Semantic HTML and CSS

4.24.1 Webpage Layout

Layout means the planned arrangement of webpage parts such as header, navigation, main content, sidebar and footer. A good layout should be planned before making the website.

Part	Purpose
Header	Contains page title, logo or main heading.
Navigation	Contains links to different pages or sections.
Main content	Contains the main information of the webpage.
Sidebar	Contains supporting links or helping elements.
Footer	Contains footer information such as copyright, contact or lower navigation.

Table 4.24: Common parts of a webpage layout

4.24.2 Methods of Creating Layout

- **Using table**: Old method; only HTML can create the structure, but it is not modern practice.
- **Using div**: Uses `div` with CSS; common for layout learning.
- **Using frameset/frame**: Old method; exam/reference only, not significant in modern webpages.
- **Using semantic HTML**: Modern method using tags such as `header`, `nav`, `main`, `section`, `article` and `footer`.

In the table below, `-` means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<div>`	`-`	`-`	Creates a block-level generic container.
	`id`	unique name	Identifies one specific container.
	`class`	class name	Groups containers for CSS styling.
	`style`	CSS declarations	Adds inline CSS to the container.
	`align`	`left`, `center`, `right`	Old alignment attribute; exam/reference only.

Tag	Attribute	Value	Meaning / Effect
	width	pixel or percentage value	Old width control; CSS is preferred.
	height	pixel or percentage value	Old height control; CSS is preferred.
	-	-	Creates an inline generic container.
	id	unique name	Identifies one specific inline part.
	class	class name	Groups inline parts for CSS styling.
	style	CSS declarations	Adds inline CSS to selected text or inline content.
<header> <nav> <main> <section> <article> <address> <footer>	-	-	Gives meaningful structure to webpage layout.
	id	unique name	Identifies a specific semantic section.
	class	class name	Applies shared CSS styling.
	style	CSS declarations	Adds inline styling when needed.
	title	text	Shows extra information as a tooltip.
	lang	language code	Identifies the language of the content.
	dir	ltr, rtl	Sets text direction for the section.
<style>	-	-	Stores internal CSS rules inside the HTML document.
	type	text/css	Old explicit CSS type value; usually unnecessary in HTML5.
<link>	rel	stylesheet	Tells the browser the linked file is a stylesheet.
	href	file path or URL	Gives the CSS file location.
	type	text/css	Old explicit CSS type value; usually unnecessary in HTML5.

Table 4.25: Main tags, attributes and values for containers, semantic HTML and CSS

4.24.3 Div and Span

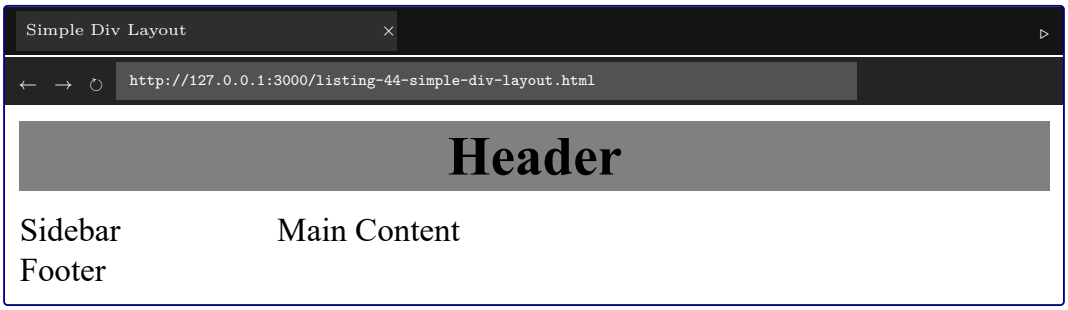
- `div` is a block-level container used to divide a webpage into sections.
- `span` is an inline container used to mark a small part of text.

```
<div class="container" style="width:100%;">
    <div class="header" style="background-color:gray;">
        <h1 align="center">Header</h1>
    </div>
    <div class="sidebar" style="float:left; width:25%;">
        Sidebar
    </div>
```

```html
    <div class="content" style="float:left; width:75%;">
        Main Content
    </div>
    <div class="footer" style="clear:left;">
        Footer
    </div>
</div>
```

Listing 4.44: Simple Div Layout

Browser Output

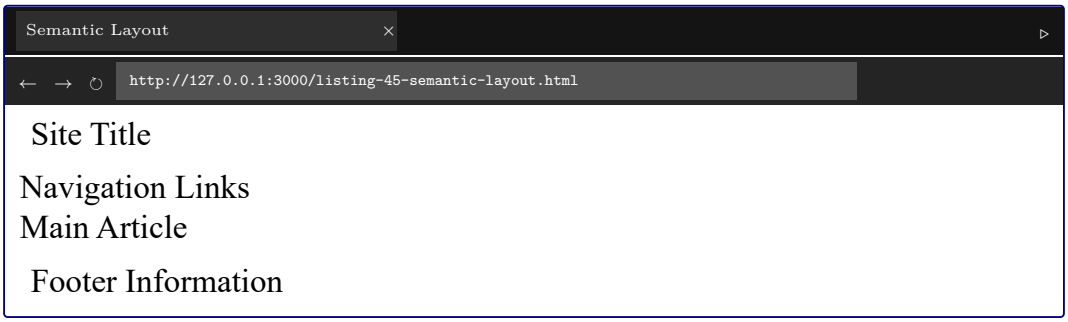


4.24.4 Semantic Layout Tags

```html
<header>Site Title</header>
<nav>Navigation Links</nav>
<main>
    <section>
        <article>Main Article</article>
    </section>
</main>
<footer>Footer Information</footer>
```

Listing 4.45: Semantic Layout

Browser Output



4.24.5 CSS Basics

CSS means Cascading Style Sheets. HTML creates the structure of a page, and CSS controls the presentation such as color, font, spacing, border and layout.

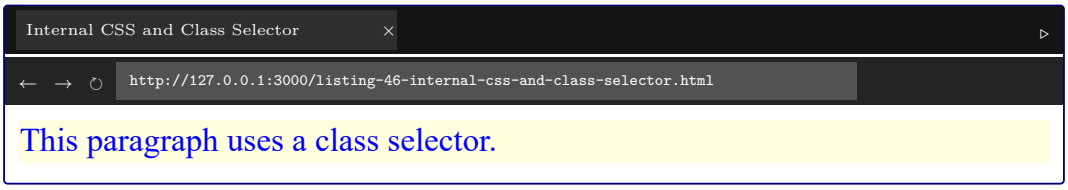
- **Inline CSS**: written inside the `style` attribute of a tag.
- **Internal CSS**: written inside a `style` tag in the `head` section.
- **External CSS**: written in a separate CSS file and connected with the `link` tag.
- Class selectors begin with a dot, such as `.notice`.

▷ ID selectors begin with a hash sign, such as `#main`.

```html
<head>
    <style>
        .notice {
            color: blue;
            background-color: lightyellow;
        }
    </style>
</head>
<body>
    <p class="notice">This paragraph uses a class selector.</p>
</body>
```

Listing 4.46: Internal CSS and Class Selector

Browser Output



Summary

Use semantic tags for meaningful page structure, `div`/`span` for generic containers, and CSS for visual design.

4.25 HTML Forms

HTML Form

An HTML form is an area of a webpage that allows users to enter or select data.

Forms are useful for understanding dynamic websites because users submit data through forms. A form can contain many controls, but one `form` element should not contain another `form` element.

4.25.1 Form Tags, Attributes and Values

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<form>`	-	-	Defines the form area.
	`action`	file path or URL	Specifies where submitted form data will be sent.
	`method`	`get`, `post`	Specifies how form data will be sent.
	`name`	form name	Gives a name to the form.
	-	-	Creates an input control.
		`text`	Creates a single-line text field.

Tag	Attribute	Value	Meaning / Effect
		password	Creates a password field that hides typed characters.
		radio	Creates a radio button for one choice from a group.
		checkbox	Creates a checkbox for one or more choices.
		submit	Creates a submit button.
		button	Creates a normal command button.
	name	field name	Names the input so submitted data can be identified.
	value	text value	Sets the submitted value or button text.
	placeholder	hint text	Shows a hint inside a text field.
	checked	-	Selects a radio button or checkbox by default.
<label>	-	-	Gives a text label for an input control.
	for	input id	Connects the label with a specific input.
	id	unique name	Identifies the label if needed.
<fieldset>	-	-	Groups related form controls.
	name	group name	Names the group of controls.
	disabled	-	Disables the grouped controls.
<legend>	-	-	Gives a title to a fieldset.
<select>	-	-	Creates a drop-down list.
	name	field name	Names the selected data.
	size	number	Sets how many options are visible.
	multiple	-	Allows more than one option to be selected.
<option>	-	-	Defines one choice inside a select list.
	value	option value	Sets the submitted value for the option.
	selected	-	Selects the option by default.
<optgroup>	-	-	Groups related options inside a select list.
	label	group label	Shows the group title in the list.
<textarea>	-	-	Creates a multi-line text input area.
	name	field name	Names the textarea data.
	rows	number	Controls textarea height.
	cols	number	Controls textarea width.

Tag	Attribute	Value	Meaning / Effect
`<button>`	-	-	Creates a clickable button.
	`type`	`button`, `submit`, `reset`	Sets the button behavior.
	`name`	button name	Names the button if submitted or scripted.

Table 4.26: Main tags, attributes and values for HTML forms

4.25.2 Basic Form Structure

```html
<form action="submit.html" method="post">
    <label>Name:</label>
    <input type="text" name="student_name">
    <input type="submit" value="Send">
</form>
```

Listing 4.47: Basic Form Structure

Browser Output

What to notice

- `form` groups the input controls.
- `action` gives the destination that receives submitted data.
- `method` commonly uses `get` or `post`.
- HTML creates the form, but another program processes the submitted data.

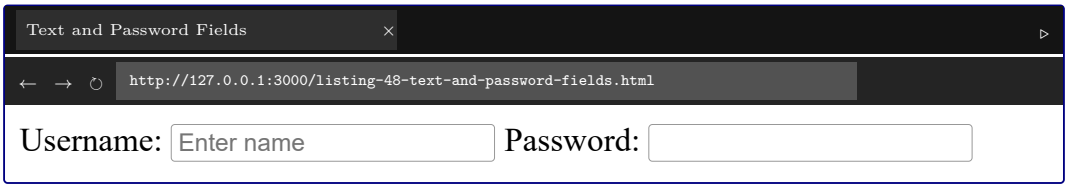
4.25.3 Text and Password Inputs

Text input accepts one line of normal text. Password input hides the typed characters on the screen.

```html
<label>Username:</label>
<input type="text" name="username" placeholder="Enter name">

<label>Password:</label>
<input type="password" name="password">
```

Listing 4.48: Text and Password Fields

Browser Output



What to notice

- `type="text"` creates a normal text box.
- `type="password"` hides typed characters.
- `placeholder` gives a hint inside the box.

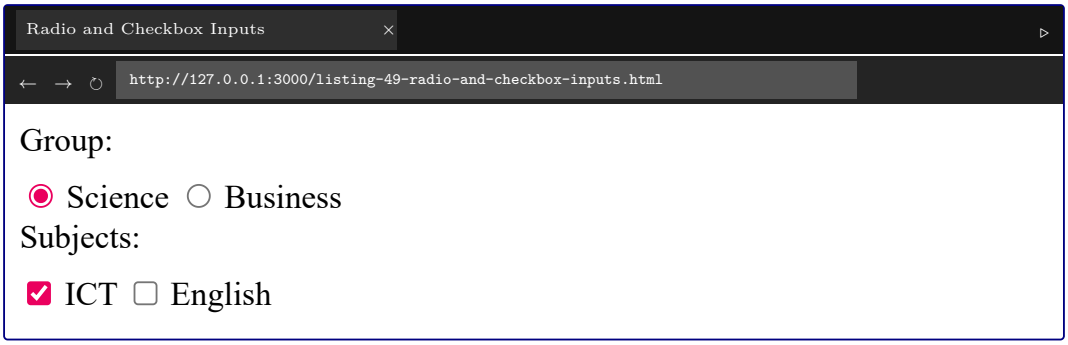
4.25.4 Radio Buttons and Checkboxes

Radio buttons are used when only one option should be selected from a group. Checkboxes are used when one or more options may be selected.

```html
<p>Group:</p>
<input type="radio" name="group" value="science" checked> Science
<input type="radio" name="group" value="business"> Business

<p>Subjects:</p>
<input type="checkbox" name="subject" value="ict" checked> ICT
<input type="checkbox" name="subject" value="english"> English
```

Listing 4.49: Radio and Checkbox Inputs

Browser Output



What to notice

- Radio buttons with the same `name` form one group.
- Only one radio button in the same group can be selected.
- More than one checkbox can be selected.
- `checked` selects an option by default.

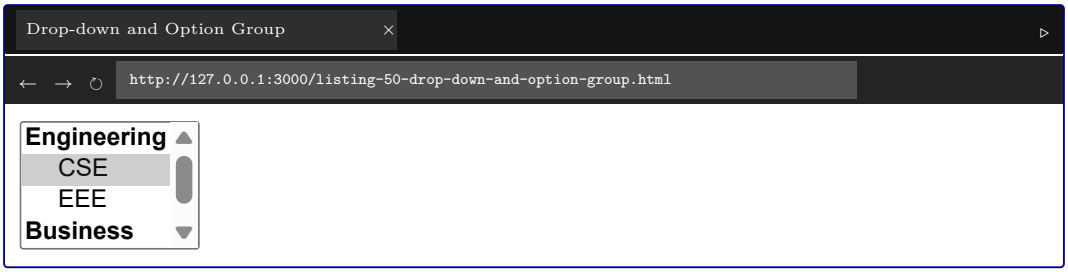
4.25.5 Select, Option and Optgroup

A `select` element creates a drop-down list. `option` defines each choice, and `optgroup` groups related choices.

```html
<select name="department" size="4" multiple>
    <optgroup label="Engineering">
        <option value="cse" selected>CSE</option>
        <option value="eee">EEE</option>
    </optgroup>
    <optgroup label="Business">
        <option value="bba">BBA</option>
    </optgroup>
</select>
```

Listing 4.50: Drop-down and Option Group

Browser Output



What to notice

- `selected` chooses an option by default.
- `multiple` allows more than one option to be selected.
- `size` controls how many options are visible.

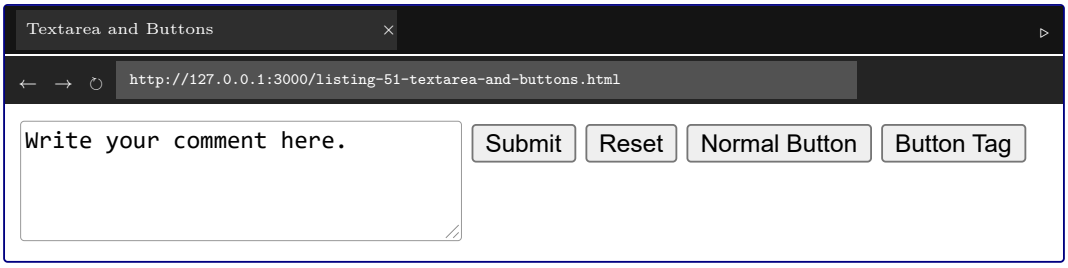
4.25.6 Textarea and Buttons

`textarea` is used for multi-line input. Buttons are used to submit, reset or trigger commands.

```
<textarea name="comment" rows="4" cols="30">
Write your comment here.
</textarea>

<input type="submit" value="Submit">
<input type="reset" value="Reset">
<input type="button" value="Normal Button">
<button type="button">Button Tag</button>
```

Listing 4.51: Textarea and Buttons

Browser Output



What to notice

- `rows` controls textarea height.
- `cols` controls textarea width.
- `submit` sends form data.
- `reset` clears the form fields.

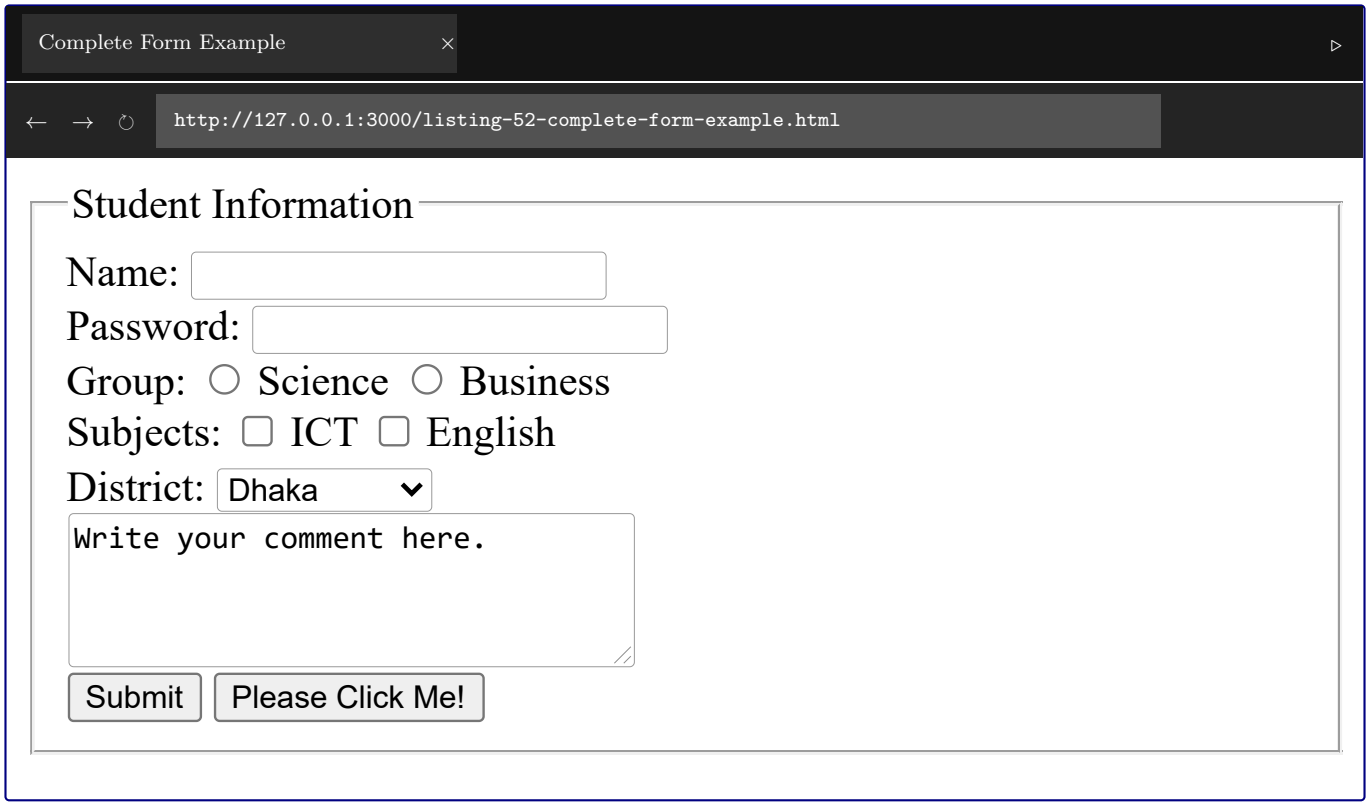
4.25.7 Form Example

```
<form action="form_action.asp" method="get">
    <fieldset>
        <legend>Student Information</legend>

        <label>Name:</label>
```

```html
        <input type="text" name="student_name"><br>

        <label>Password:</label>
        <input type="password" name="password"><br>

        <label>Group:</label>
        <input type="radio" name="group" value="science"> Science
        <input type="radio" name="group" value="business"> Business<br>

        <label>Subjects:</label>
        <input type="checkbox" name="subject" value="ict"> ICT
        <input type="checkbox" name="subject" value="english"> English<br>

        <label>District:</label>
        <select name="district">
            <option>Dhaka</option>
            <option>Chattogram</option>
        </select><br>

        <textarea name="comment" rows="4" cols="30">
Write your comment here.
        </textarea><br>

        <input type="submit" value="Submit">
        <input type="button" value="Please Click Me!">
    </fieldset>
</form>
```

Listing 4.52: Complete Form Example

Browser Output



Summary

Forms collect user input. HTML creates the form controls, but server-side processing is needed to store or process submitted data.

4.26 Code, Quote, Direction, Script and Object Tags

Some HTML tags are not used for ordinary paragraphs, links or tables. They are used to show code, keyboard input, quotations, text direction, scripts or embedded objects.

In the table below, - means no special attribute or value is required for the basic use of that tag.

Tag	Attribute	Value	Meaning / Effect
`<pre>`	-	-	Preserves spaces and line breaks exactly as written.
`<code>`	-	-	Shows computer code or program text.
`<kbd>`	-	-	Shows keyboard input typed by the user.
`<var>`	-	-	Shows a variable name.
`<dfn>`	-	-	Marks a defining instance of a term.
`<samp>`	-	-	Shows sample output from a computer program.
`<tt>`	-	-	Shows teletype-style text; exam/reference only.
`<blockquote>`	`cite`	URL	Shows a long quotation and may give its source address.
`<q>`	`cite`	URL	Shows a short inline quotation and may give its source address.
`<cite>`	-	-	Shows the title of a cited work or reference.
`<abbr>`	`title`	full form text	Marks an abbreviation and gives its full form.
`<acronym>`	`title`	full form text	Old acronym tag; exam/reference only.
`<bdo>`	`dir`	`ltr`	Forces text direction from left to right.
	`dir`	`rtl`	Forces text direction from right to left.
`<script>`	-	-	Adds script code inside the HTML document.
	`src`	file path or URL	Connects an external script file.
	`type`	script MIME type	Old explicit script type; usually unnecessary in HTML5.

Tag	Attribute	Value	Meaning / Effect
	defer	-	Delays script execution until the document is parsed.
<noscript>	-	-	Shows fallback content when scripting is unavailable.
<object>	data	file path or URL	Gives the embedded object source.
	type	MIME type	Identifies the embedded resource type.
	width	pixel value	Sets object width.
	height	pixel value	Sets object height.
<param>	name	parameter name	Gives the parameter name for an object.
	value	parameter value	Gives the value of that parameter.
<embed>	src	file path or URL	Embeds external media or plugin content.
	type	MIME type	Identifies the embedded resource type.
	width	pixel value	Sets embedded content width.
	height	pixel value	Sets embedded content height.
<applet>	code	class file name	Old Java applet source; exam/reference only.
	width	pixel value	Sets old applet width.
	height	pixel value	Sets old applet height.
	alt	text	Shows alternative text if the applet cannot run.

Table 4.27: Main tags, attributes and values for code, quote, direction, script and object content

4.26.1 Code and Computer Output Tags

```
<pre>
Line one
    Line two with spaces
</pre>
<p>Use <code>document.write()</code> in script examples.</p>
<p>Press <kbd>Ctrl</kbd> + <kbd>S</kbd> to save.</p>
<p><var>x</var> is a variable.</p>
<p><samp>Hello World</samp></p>
```

Listing 4.53: Code and Output Tags

Browser Output

Code and Output Tags

http://127.0.0.1:3000/listing-53-code-and-output-tags.html

```
Line one
    Line two with spaces
```

Use `document.write()` in script examples.

Press `Ctrl` + `S` to save.

x is a variable.

4.26.2 Quote, Citation and Direction Tags

```html
<blockquote>This is a long quotation.</blockquote>
<p>HTML means <q>HyperText Markup Language</q>.</p>
<p><abbr title="World Wide Web">WWW</abbr></p>
<p><cite>ICT Book</cite></p>
<p><bdo dir="rtl">This text is shown right to left.</bdo></p>
```

Listing 4.54: Quote and Direction Tags

Browser Output

Quote and Direction Tags

http://127.0.0.1:3000/listing-54-quote-and-direction-tags.html

This is a long quotation.

HTML means “HyperText Markup Language”.

WWW

4.26.3 Script, Noscript, Object and Applet

```html
<script>
    document.write("Hello from JavaScript");
</script>
<noscript>Your browser does not support JavaScript.</noscript>

<object data="notice.pdf" width="400" height="300">
    <param name="autoplay" value="false">
</object>
```

Listing 4.55: Script and Object Tags

Browser Output

Script and Object Tags

http://127.0.0.1:3000/listing-55-script-and-object-tags.html

Hello from JavaScript

What to notice

- HTML can connect JavaScript, but JavaScript does the programming work.
- `noscript` is fallback content.

▷ `applet`, many `embed` uses and `tt` are old exam/reference items, not modern practice.

4.27 Integrated Mini Project

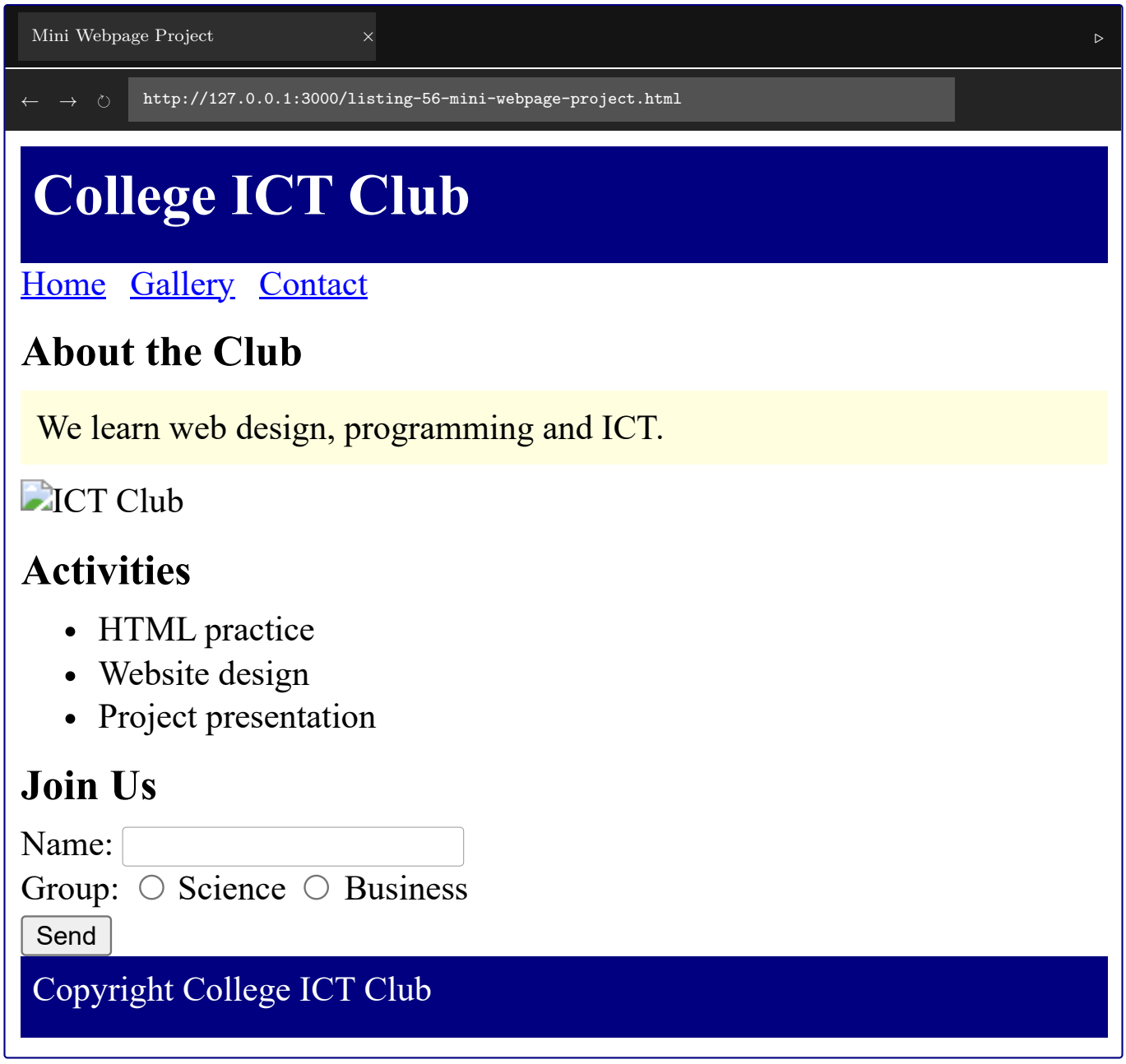
This final example combines the main ideas from the HTML lessons. It uses document structure, semantic layout, navigation links, an image, a list, a form and simple CSS in one small webpage.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>College ICT Club</title>
    <style>
        body { font-family: Arial; }
        header, footer { background-color: navy; color: white; }
        nav a { color: blue; margin-right: 10px; }
        .notice { background-color: lightyellow; padding: 8px; }
    </style>
</head>
<body>
    <header>
        <h1>College ICT Club</h1>
    </header>

    <nav>
        <a href="index.html">Home</a>
        <a href="gallery.html">Gallery</a>
        <a href="contact.html">Contact</a>
    </nav>

    <main>
        <section>
            <h2>About the Club</h2>
            <p class="notice">We learn web design, programming and ICT.</p>
            <img src="club.jpg" alt="ICT Club" width="250">
        </section>

        <section>
            <h2>Activities</h2>
            <ul>
                <li>HTML practice</li>
                <li>Website design</li>
                <li>Project presentation</li>
            </ul>
        </section>

        <section>
            <h2>Join Us</h2>
            <form action="join.php" method="post">
                <label>Name:</label>
                <input type="text" name="student_name"><br>
                <label>Group:</label>
                <input type="radio" name="group" value="science"> Science
```

```html
                <input type="radio" name="group" value="business"> Business<br>
                <input type="submit" value="Send">
            </form>
        </section>
    </main>

    <footer>
        <p>Copyright College ICT Club</p>
    </footer>
</body>
</html>
```

Listing 4.56: Mini Webpage Project

Browser Output



What to notice

- The page has a correct HTML document skeleton.
- Semantic tags organize the page into header, navigation, main content, sections and footer.

- CSS controls presentation, while HTML controls structure.
- The form collects data, but server-side code is required to process the submitted information.

4.28 Quick Revision Checklist

- Do not confuse **Internet** and **World Wide Web**: the Internet is the global network infrastructure; the Web is a service that runs on it using webpages, browsers, links and web servers.
- Do not confuse **web browser** and **search engine**: a browser displays webpages from a URL; a search engine finds webpages when the exact URL is unknown.
- In web access, the order is usually: user enters URL, browser reads the domain, DNS finds the server IP address, HTTP/HTTPS requests the file, server responds, browser renders the webpage.
- **IP address**, **domain name**, **DNS**, **URL** and **hosting** are different: IP identifies the server/device, domain is the readable name, DNS connects name to IP, URL gives the full resource address, hosting stores the website files.
- A domain name must be unique and is registered through registrar systems coordinated under ICANN; hosting may contain one or many websites depending on the hosting plan.
- In a domain such as `www.dhakaboard.gov.bd`, `www` is the host/subdomain, `dhakaboard` is the SLD, `gov` is a TLD/category part, and `bd` is a country code TLD.
- Static websites show fixed prepared files; dynamic websites generate or change content using server-side scripts, user input or databases.
- HTTP transfers webpages between browser and web server; FTP is mainly for uploading and downloading files, including website files before publishing.
- HTML is a markup language, not a programming language: it structures content but does not perform logic, calculation, database work or server-side processing by itself.
- The `head` section stores document information such as title, metadata, links, styles and scripts; the `body` section contains visible webpage content.
- Paired/container tags wrap content; empty tags such as `br`, `hr`, `img`, `meta` and `link` do not require closing content.
- Many old tags and attributes are exam/reference only: `font`, `center`, `marquee`, `dir`, `menu`, `frameset`, `frame`, `applet`, `basefont` and visual attributes such as `align`, `bgcolor`, `border`, `hspace`, `vspace`.
- Modern practice is: semantic HTML for meaning, CSS for presentation, JavaScript for browser-side behavior, and server-side code/database for processing submitted or dynamic data.
- In lists, `ol` controls ordered numbering, `ul` controls bullets, `li` creates each item, and `dl`/`dt`/`dd` are used for description lists.
- In hyperlinks, `href` controls the destination, `target="_blank"` opens a new tab/window, `mailto:` starts email, and `download` suggests saving the linked file.
- In images, `src` must point to a correct file path or URL, `alt` is fallback/accessibility text, and image maps need matching `usemap`, `map name` and `area` coordinates.
- In tables, `tr` creates rows, `td`/`th` create cells, `colspan` merges columns, `rowspan` merges rows, and `cellpadding` is inside-cell space while `cellspacing` is between-cell space.
- In forms, HTML only creates input controls; `action`, `method`, `name` and `value` decide how submitted data is identified and sent, but another program must process it.

- Boolean attributes such as `checked`, `selected`, `multiple`, `controls`, `autoplay`, `loop`, `muted`, `defer` and `disabled` work by being present; they do not need separate textbook-style values.